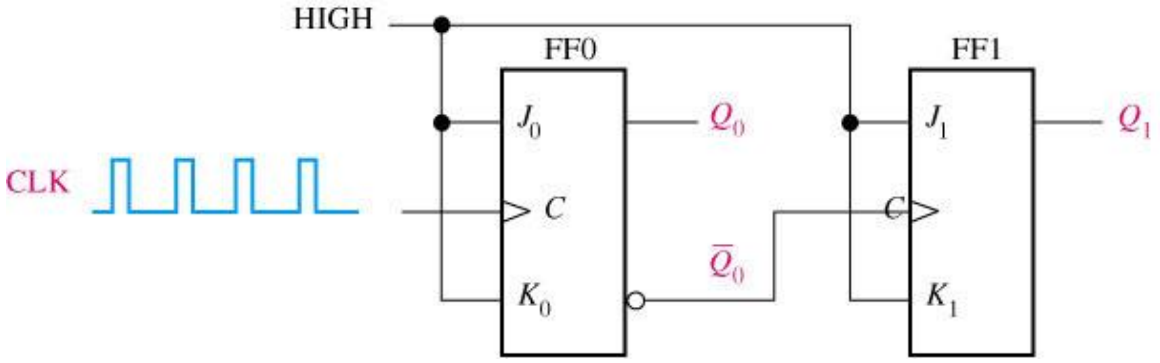
Digital Circuit and Logic Design



Chapter 8
Counters

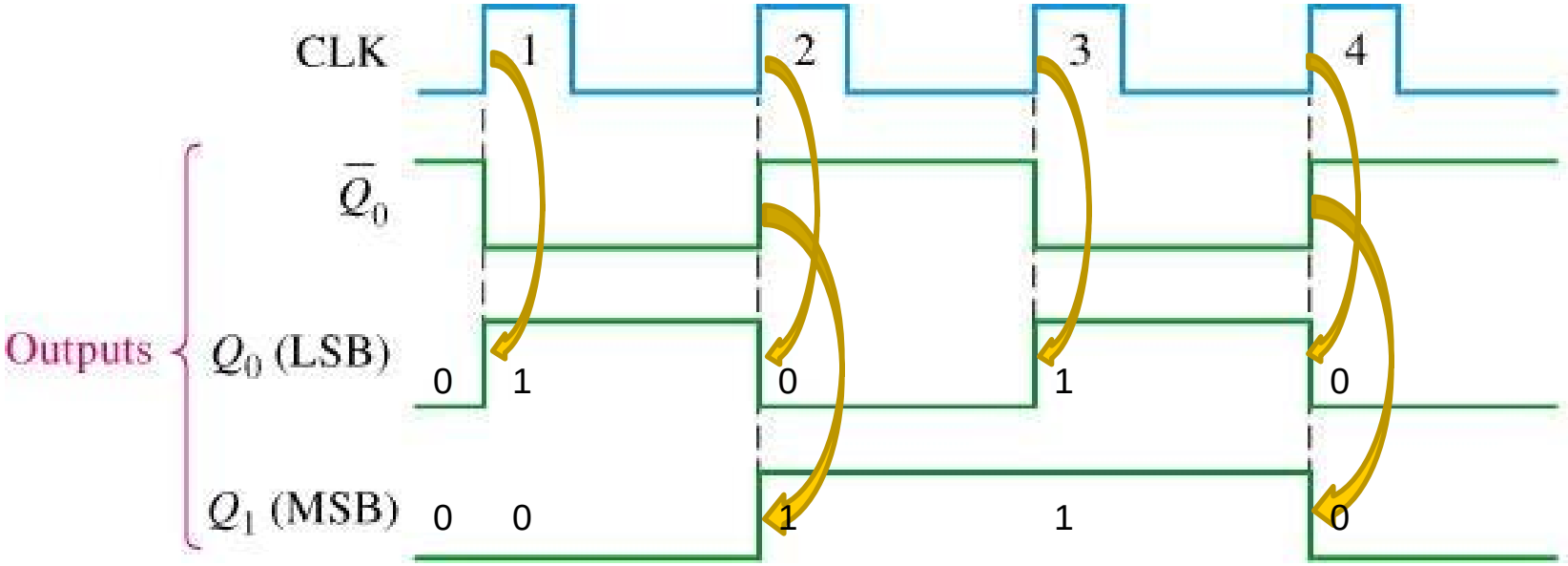
2-bit asynchronous binary counter



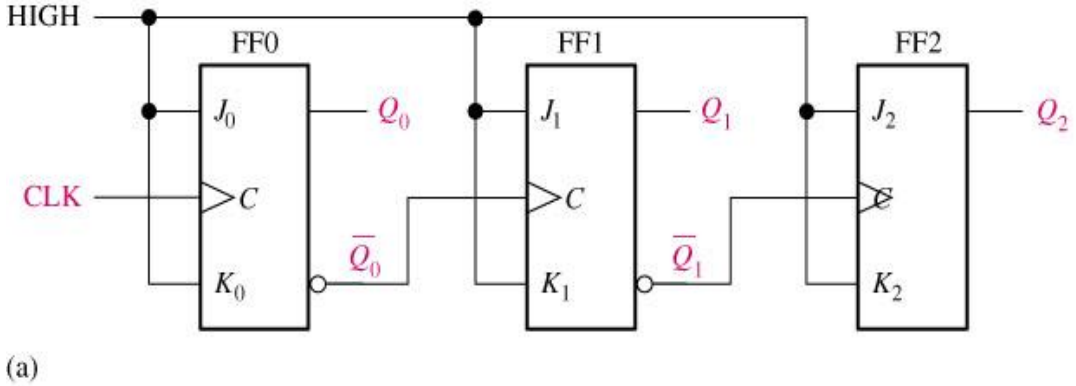
CLOCK PULSE	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4 (recycles)	0	0

*เป็นวงจร Asynchronous เนื่องจาก FF1 ไม่ถูกกระตุ้นด้วยสัญญาณ CLK

Note: การทำงานของวงจรนับ บิต ถัดไป จะเปลี่ยนสถานะก็ต่อเมื่อ บิตก่อนหน้านั้น นับเพิ่ม 1 จนเปลี่ยนเป็น 0 (ซึ่งเป็นการเปลี่ยนของขอบสัญญาณขาลง)
ดังนั้น หาก F/F ที่นำมาสร้างเป็นวงจรนับ เป็นแบบขอบขาขึ้น จำเป็นต้องใช้ Q- แทน เป็น Clock สำหรับ บิตถัดไปแทน

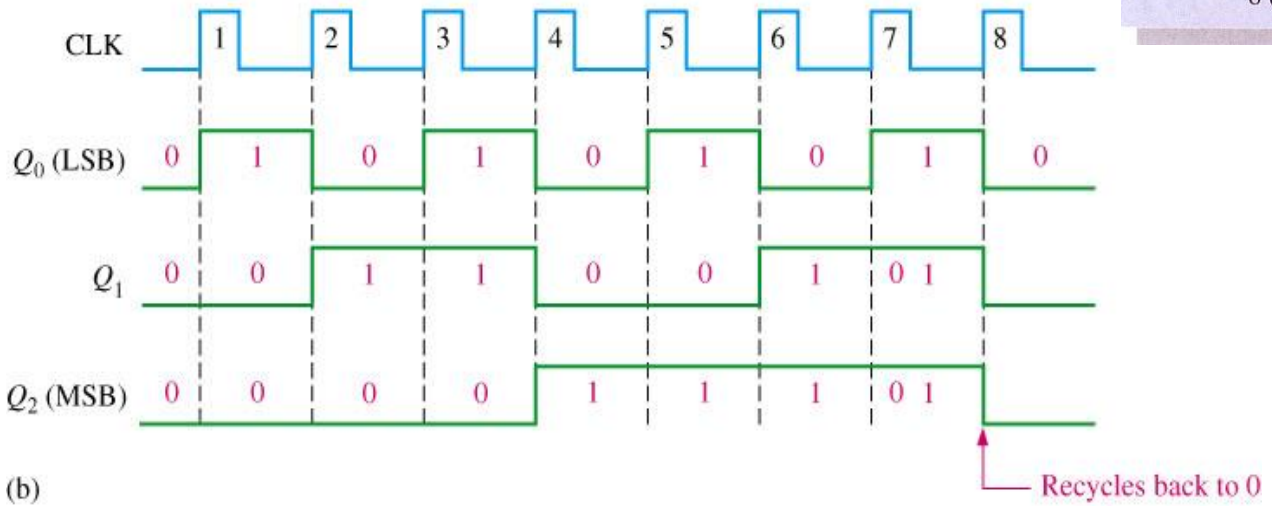


3-bit asynchronous binary counter



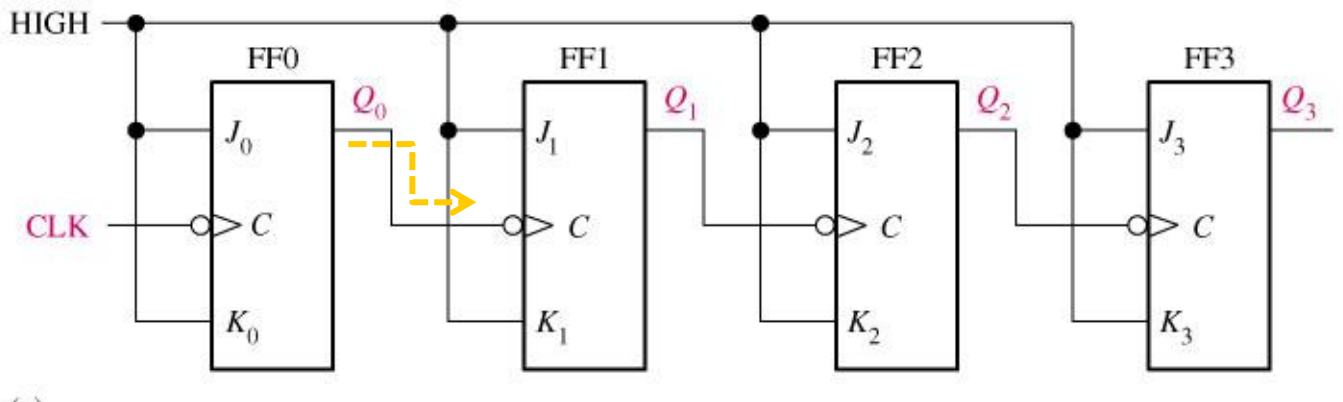
(a)

CLOCK PULSE	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8 (recycles)	0	0	0



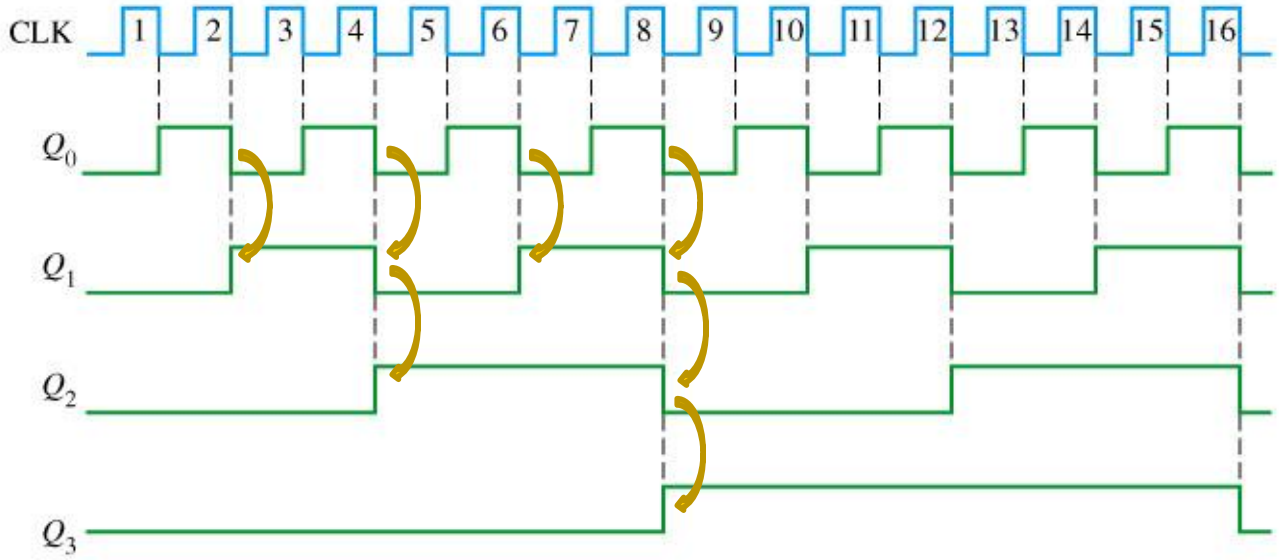
(b)

4-bit asynchronous binary counter



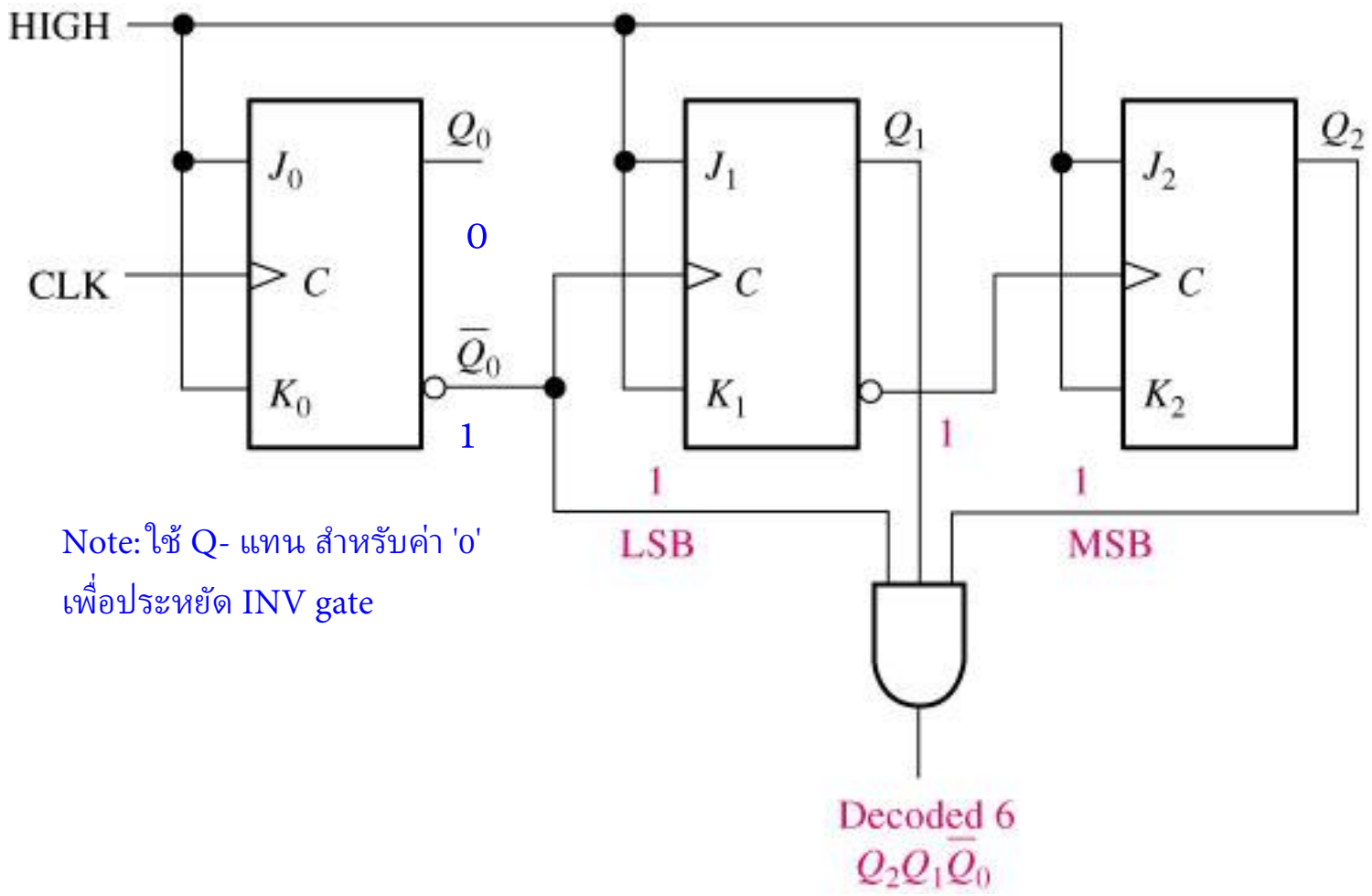
(a)

Note: FF ทำงานที่ขอบขาลง ดังนั้นต้องต่อ Q ไปยัง FF ตัวถัดไป โดยจะทำงานเมื่อ Q 1->0



(b)

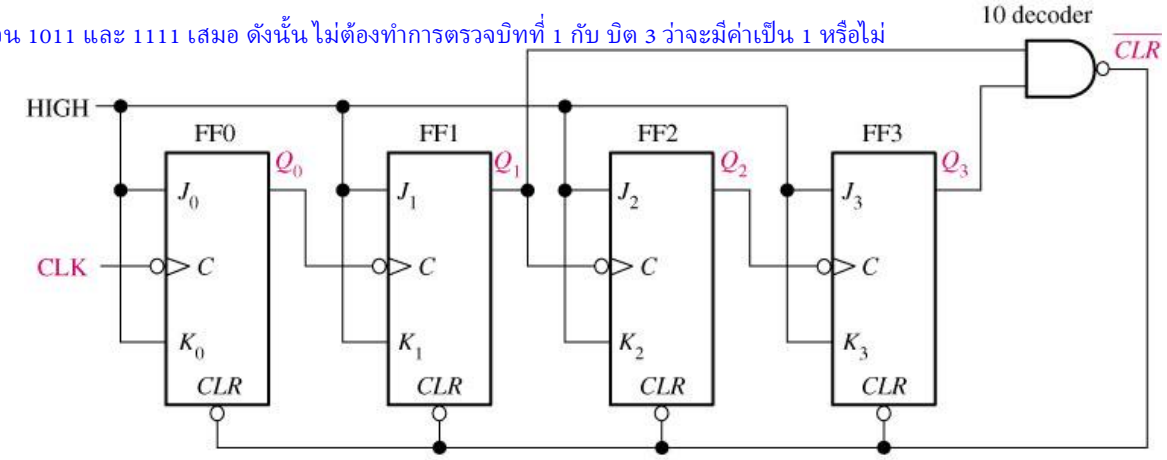
Decoding of state 6 $(110)_2$



Note: ใช้ Q- แทน สำหรับค่า '0' เพื่อประหยัด INV gate

Decade counter with asynchronous recycling

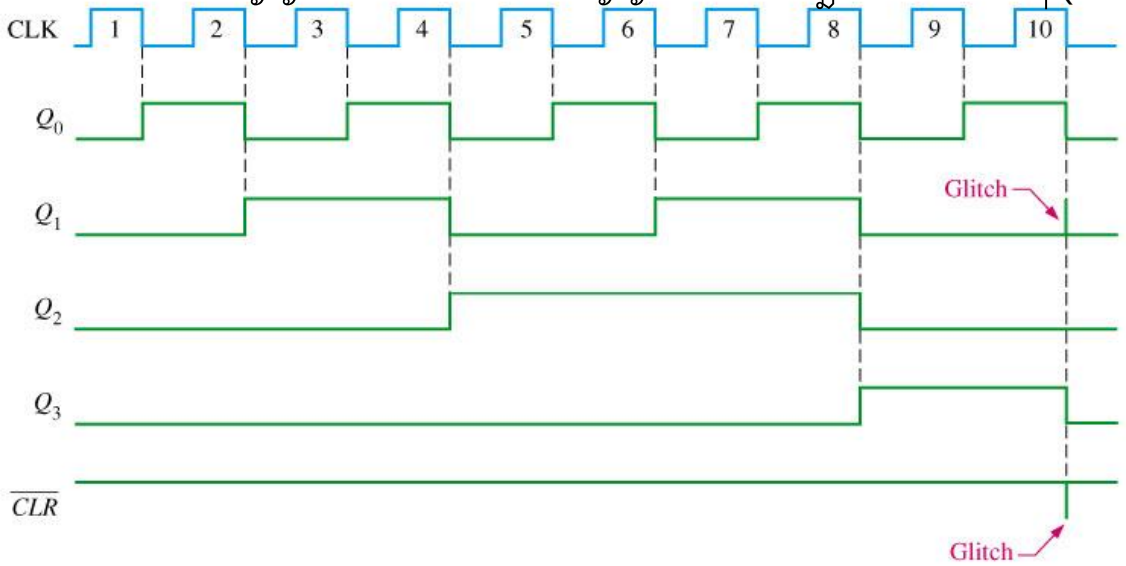
note:การนับทำให้เกิด 1010 ก่อน 1011 และ 1111 เสมอ ดังนั้นไม่ต้องทำการตรวจบิทที่ 1 กับ บิต 3 ว่าจะมีค่าเป็น 1 หรือไม่



(a)

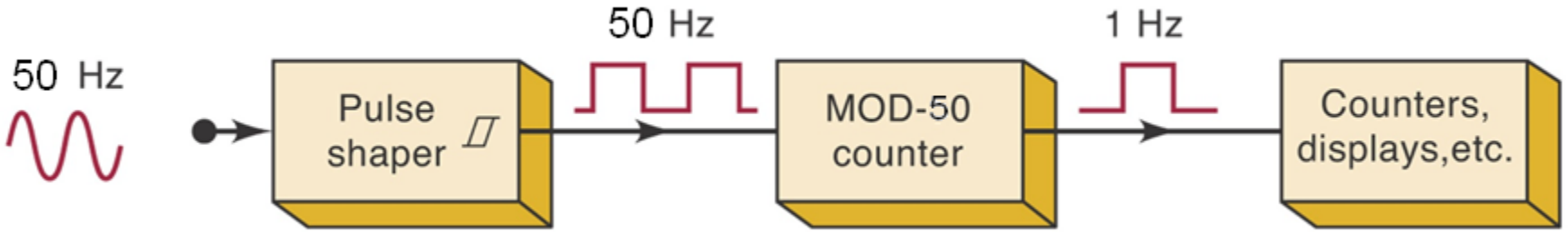
ทำการนับ pulse จนครบ 10 (= 1010_2) แล้ววนกลับไปเริ่มใหม่ (0000)

โดยสร้างสัญญาณ $\overline{CLR}$ = 0 โดยสัญญาณจะปรากฏเพียงเวลาสั้น ๆ (Glitch)



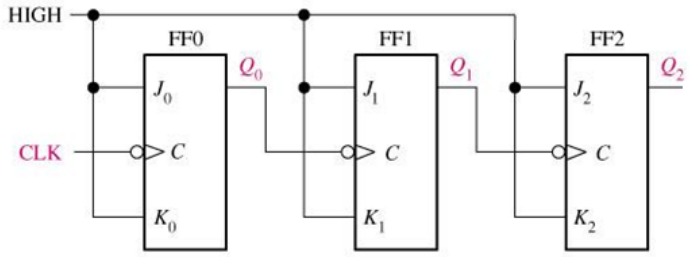
(b)

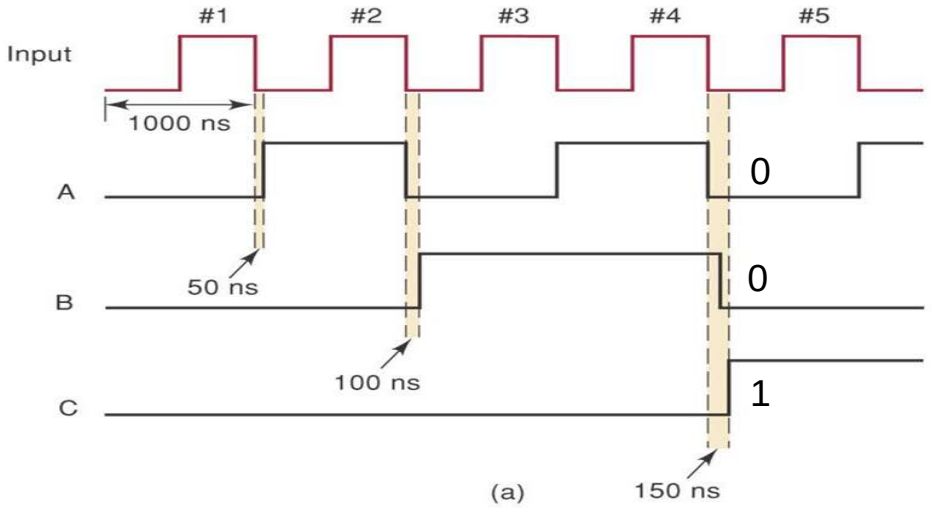
Example

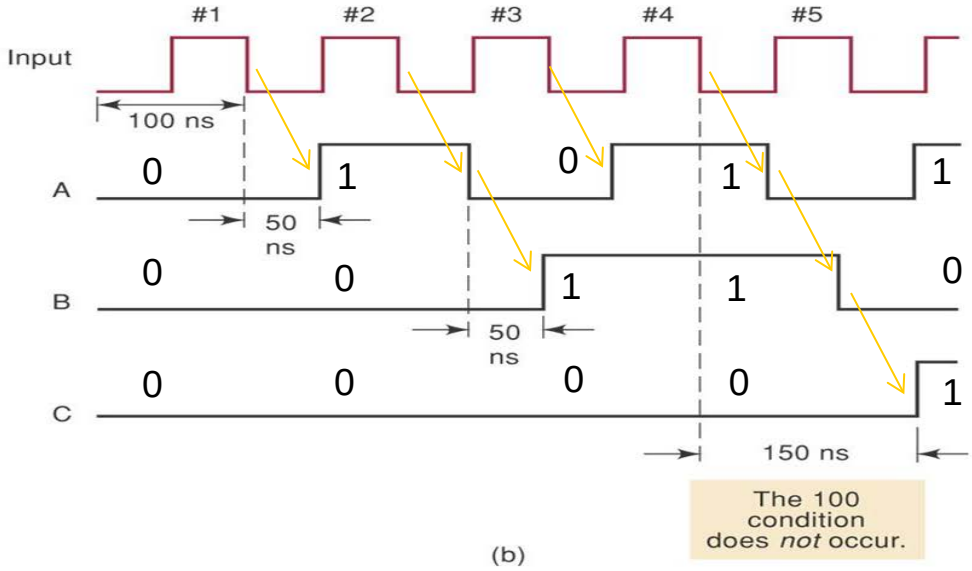


ทำการนับ pulse จนครบ 50 $(110010)_2$ แล้วสร้างสัญญาณ 1 พลัส

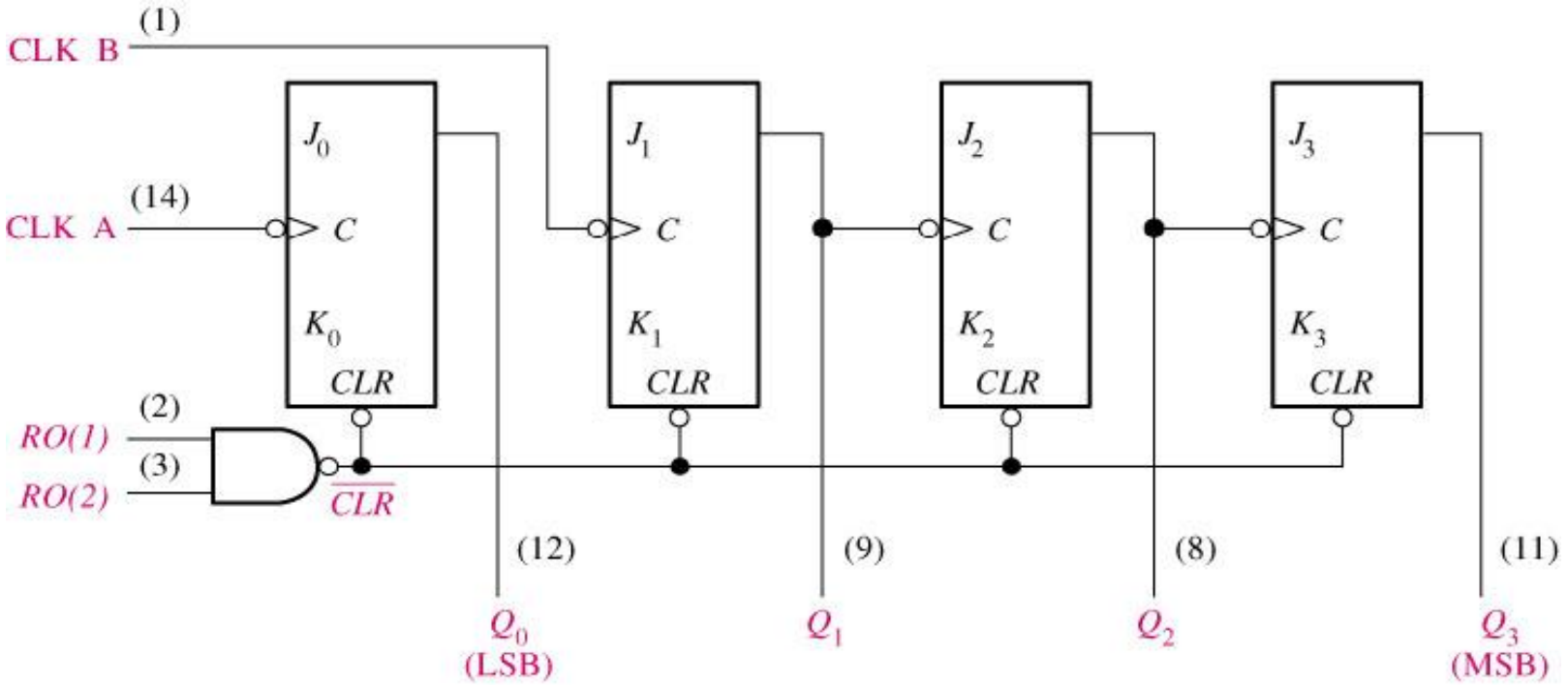
Propagation delays in a 3-bit ripple counter

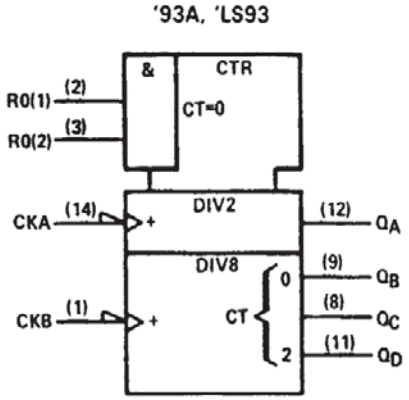




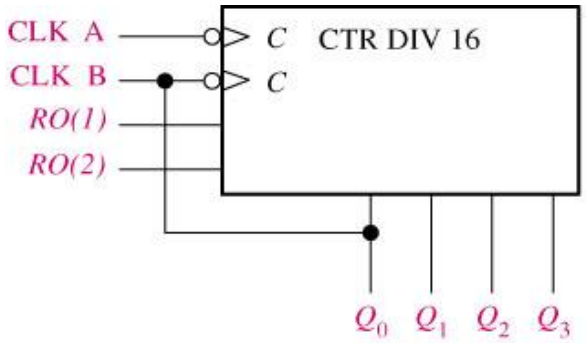


74LS93A 4-bit asynchronous binary counter 11.00฿



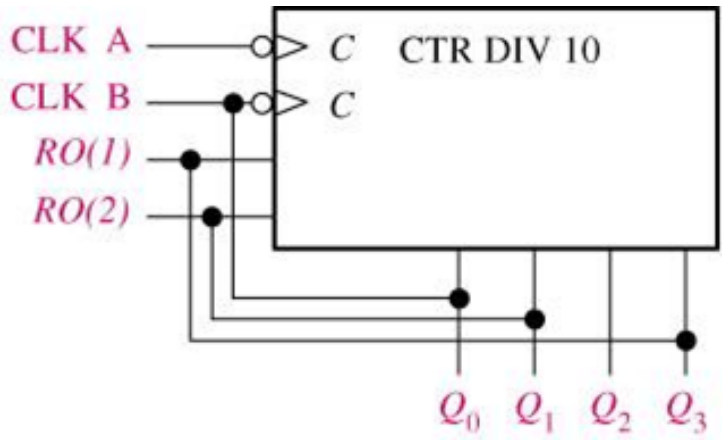


R0 : return to zero

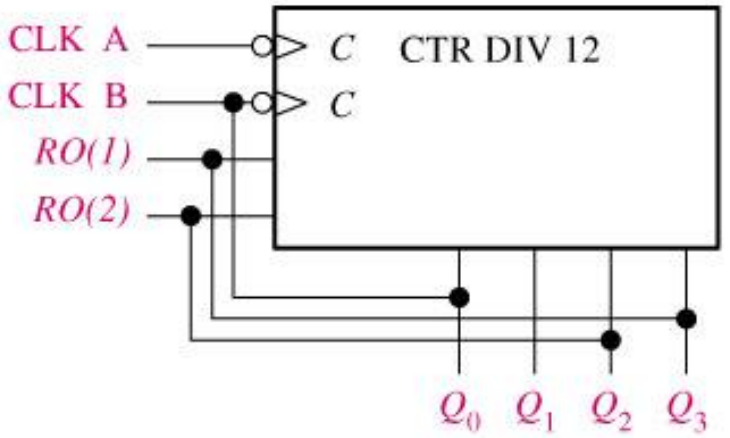


(a) 74LS93A connected as a modulus-16 counter

Configurations of the 74LS93A counter



(b) 74LS93A connected as a decade counter



$10 = 1010_2$

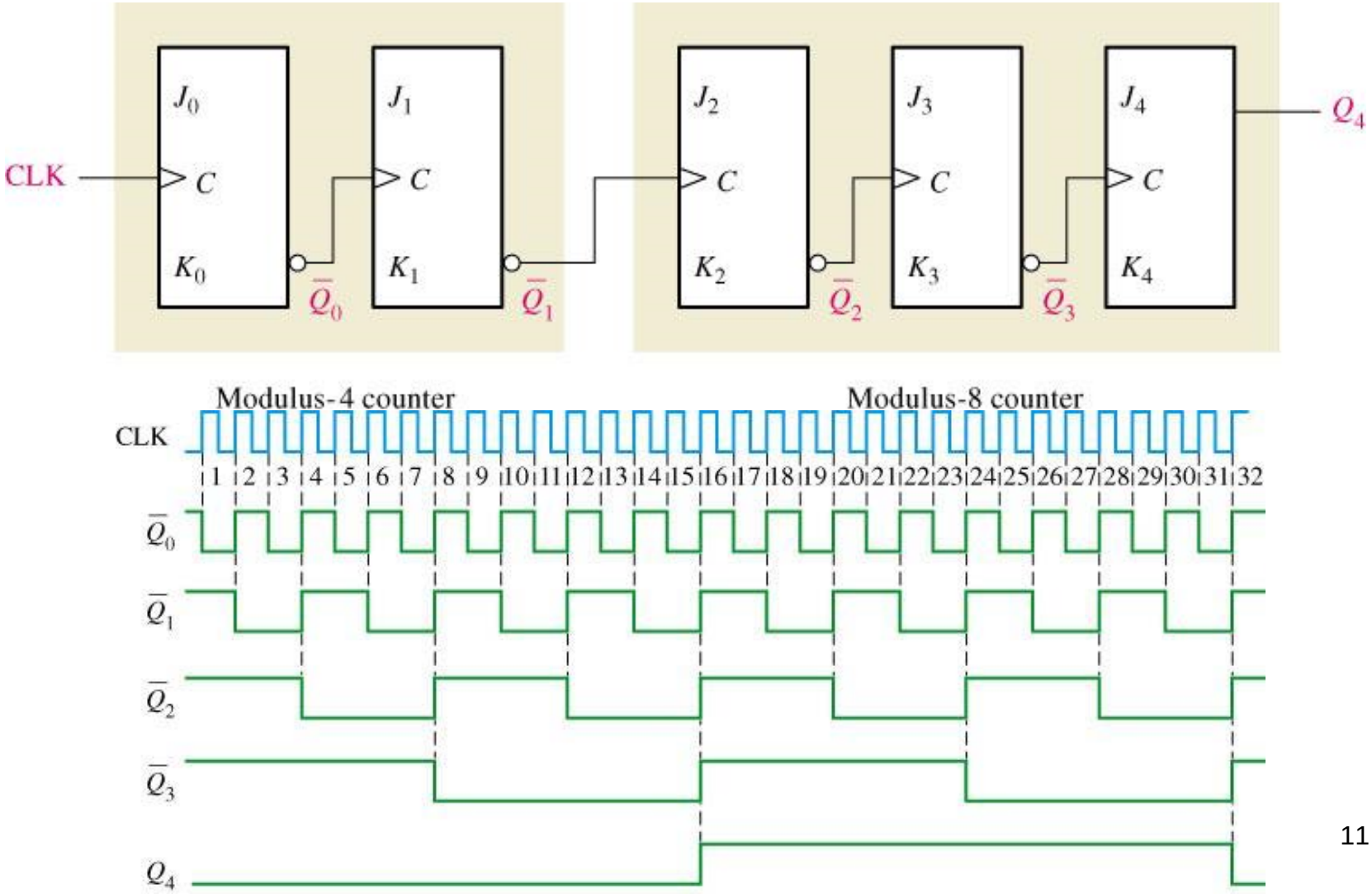
Q_3 และ $Q_1 = 1$

$12 = 1100_2$

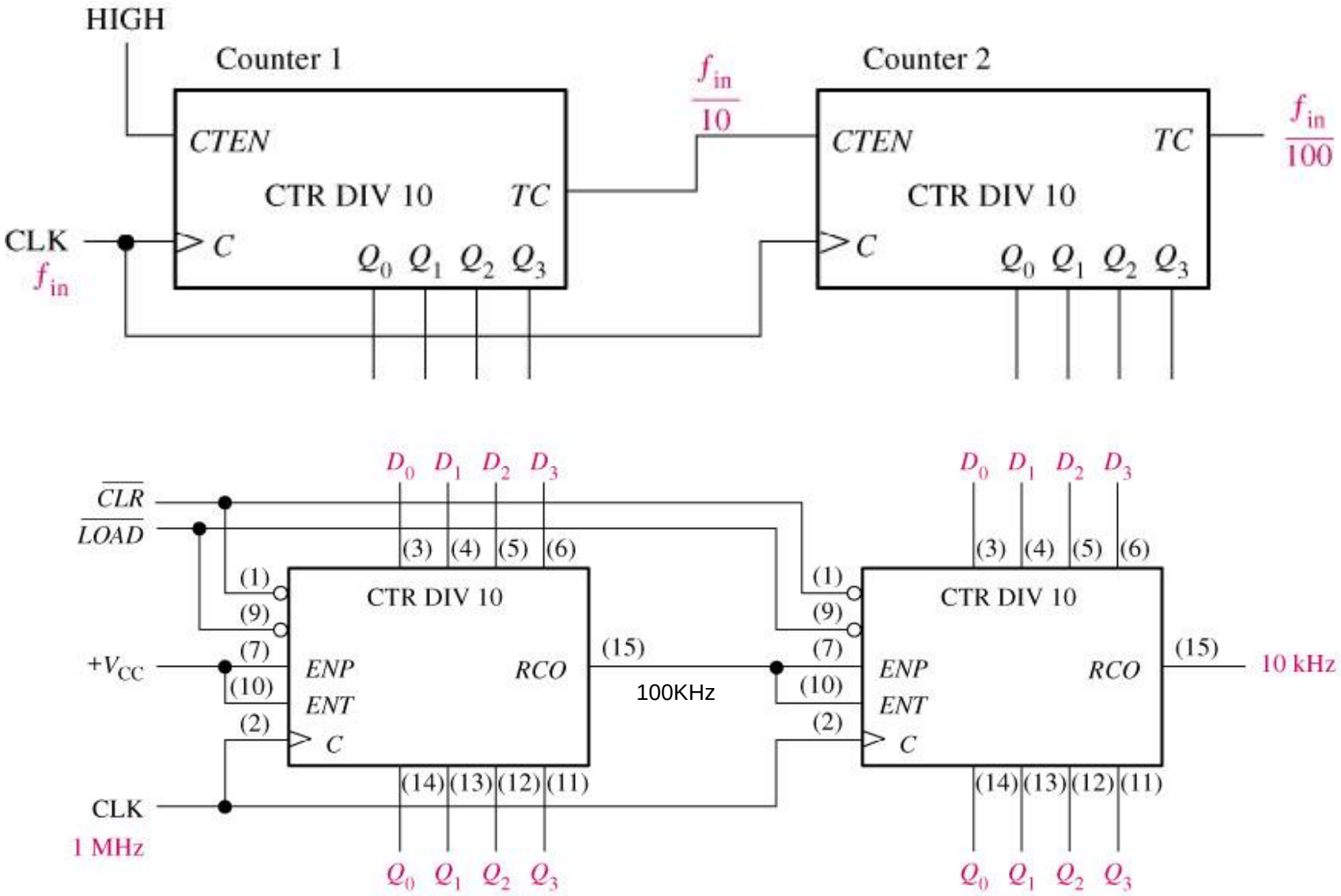
Q_3 และ $Q_2 = 1$

Two cascaded counters

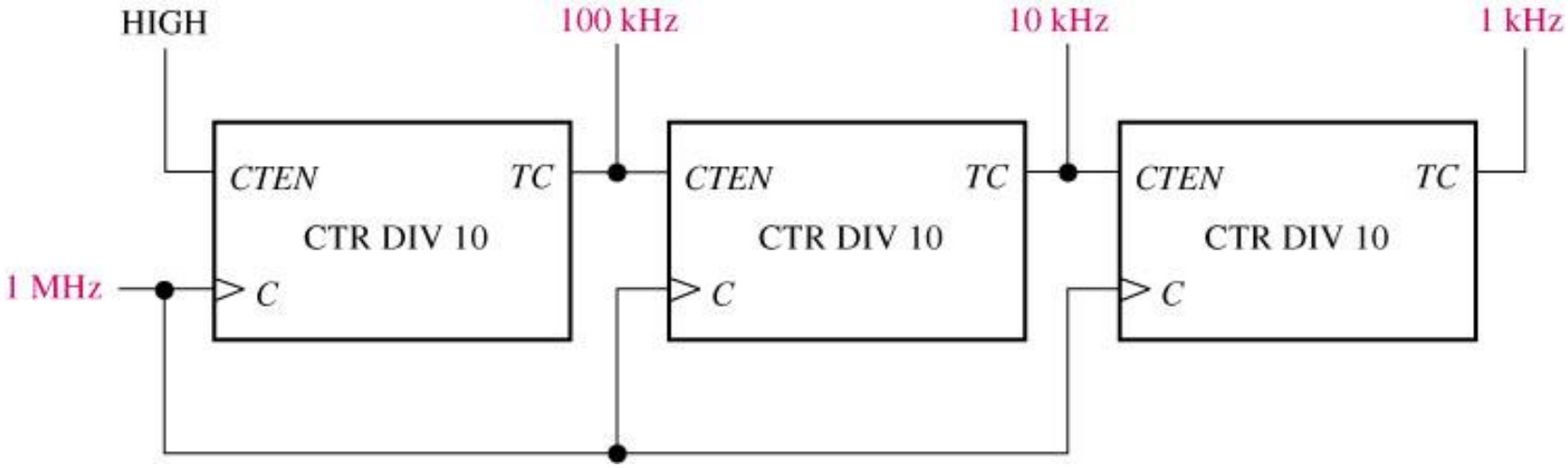
Cascaded Counters



Mod-100 counter using 2 cascaded counters

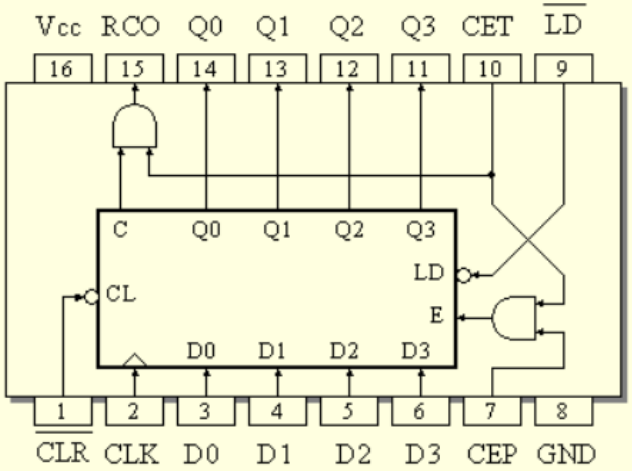


3 cascaded decade counters



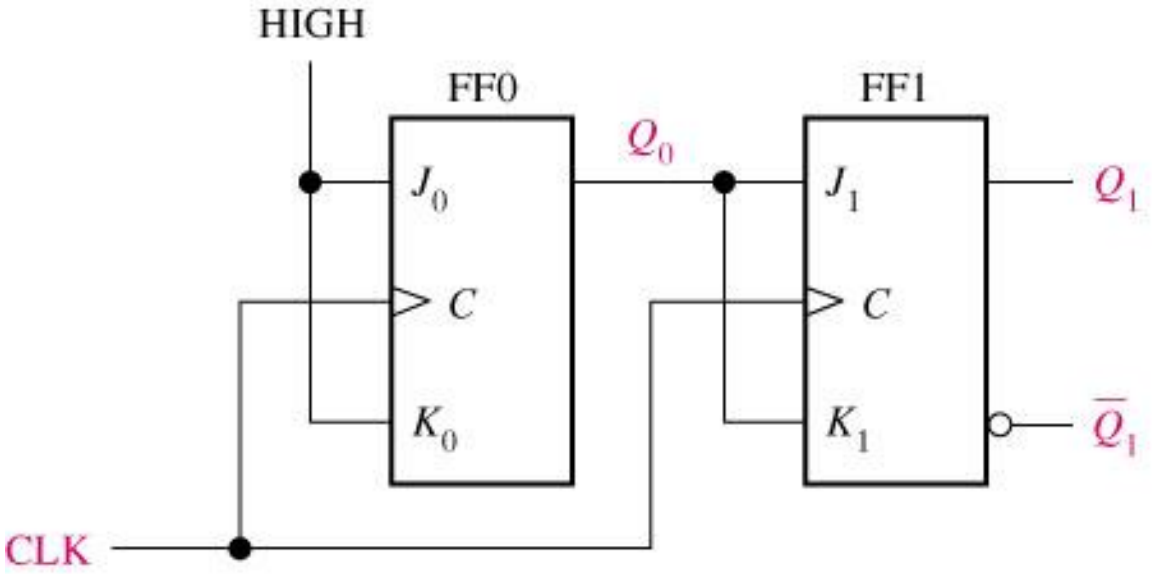
74160 Synchronous Presettable 4-Bit BCD Counter with Async. Clear

Pin Arrangement

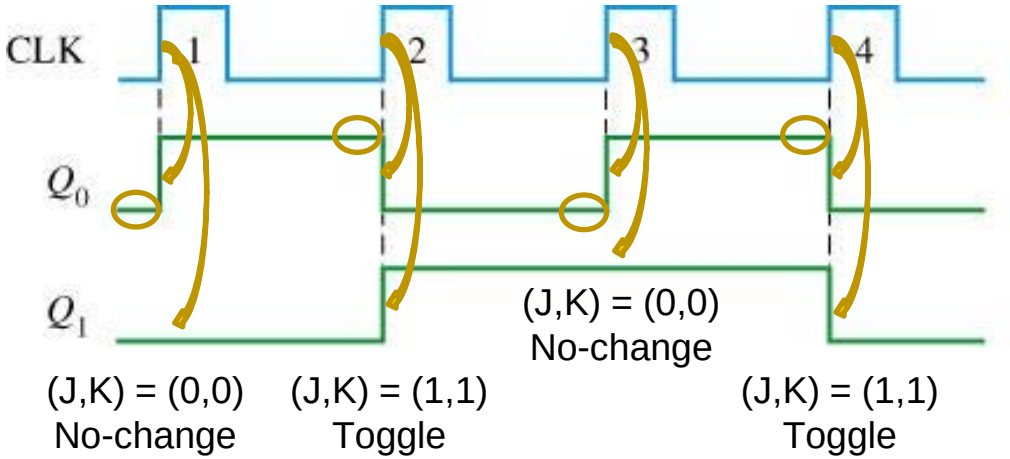


CLK	Rising Edge Trigger
CEP	Count Enable Parallel
CET	Count Enable Trickle
RCO	Ripple Carry Out
CLR'	Asynchronous Clear
LD'	Synchronous Load

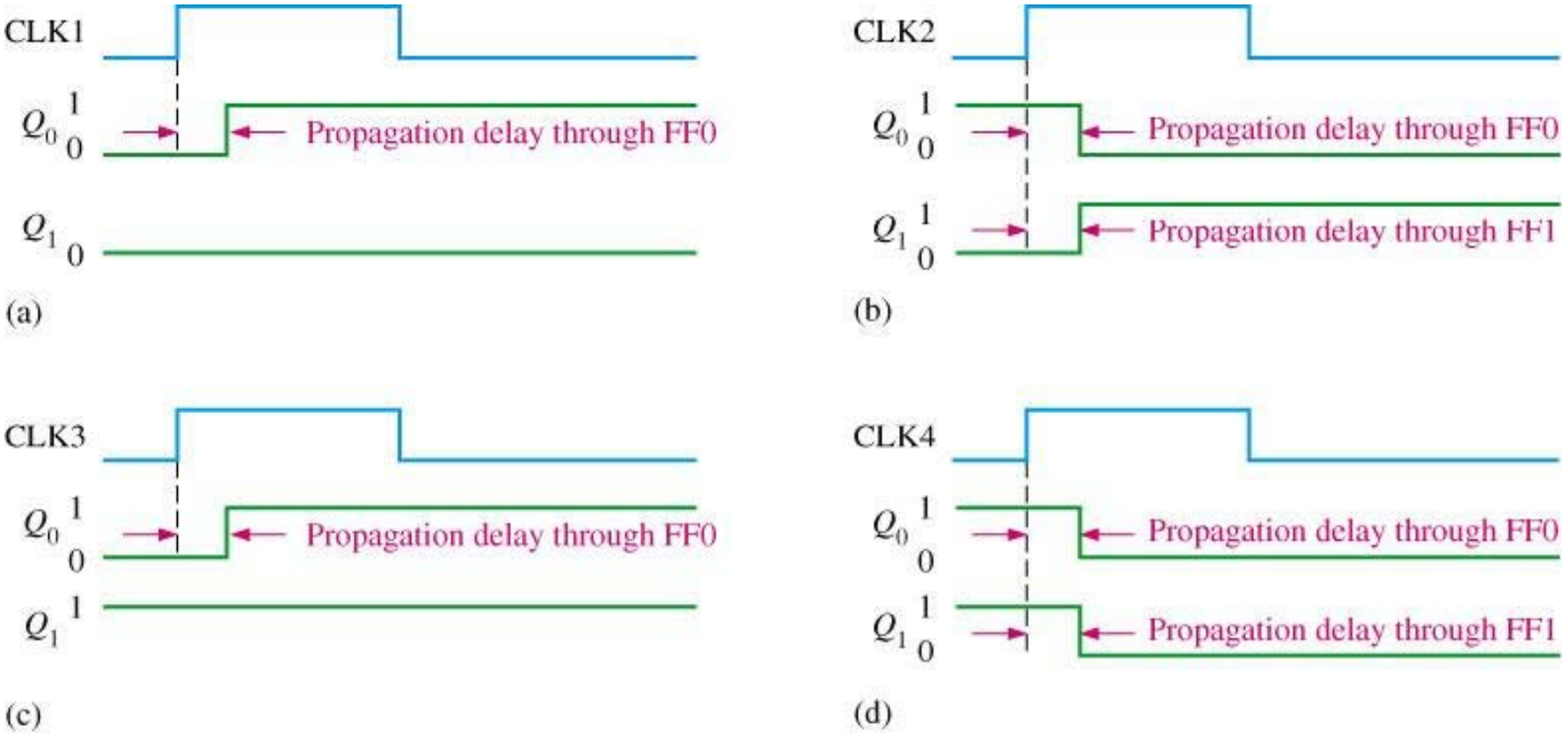
2-bit synchronous binary counter



เป็นวงจร Synchronous เนื่องจาก FF ทุกตัวถูกกระตุ้นให้ทำงานพร้อมกันด้วยสัญญาณ CLK

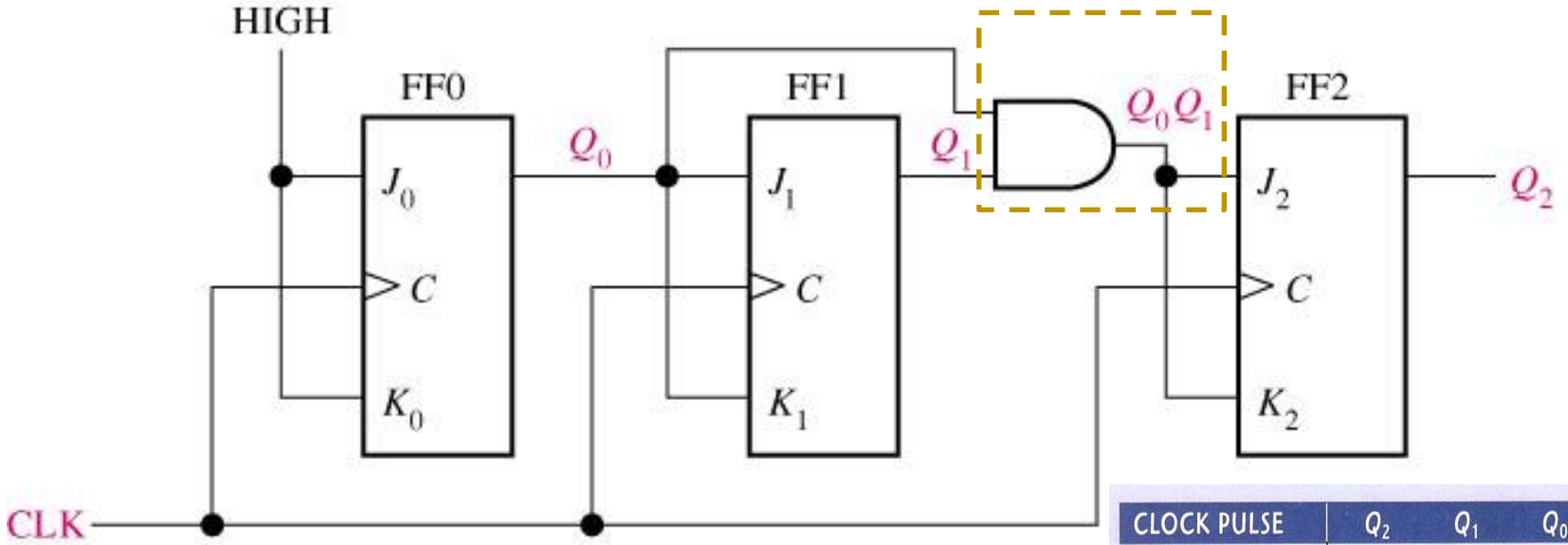


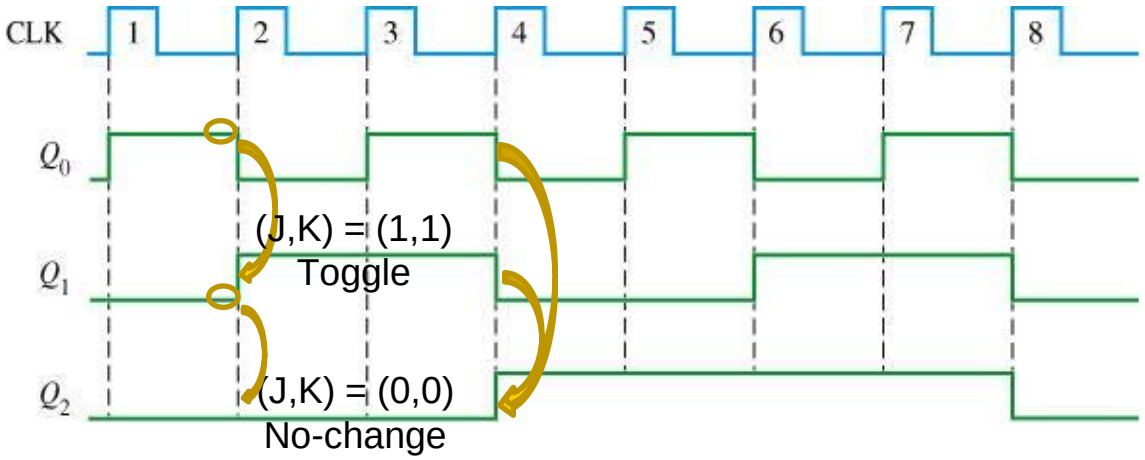
2-bit synchronous counter operation



Propagation delay จะไม่ถูกส่งต่อเหมือนวงจร Asynchronous เนื่องจาก FF ทุกตัวทำงานพร้อมกันตามสัญญาณ CLK

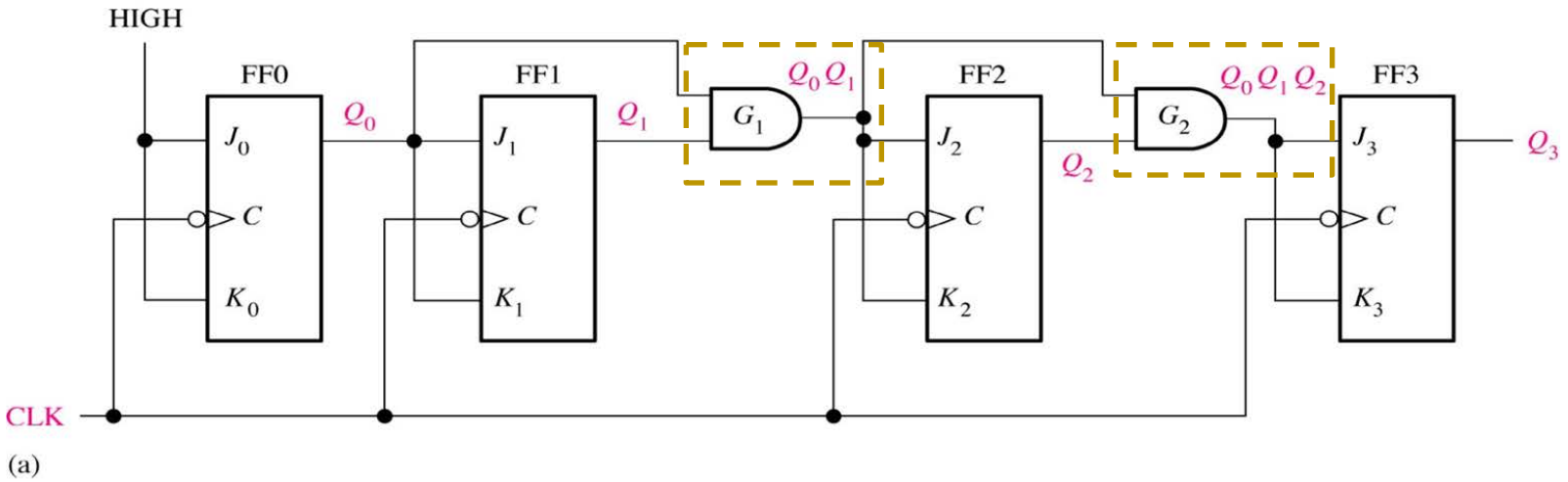
3-bit synchronous binary counter



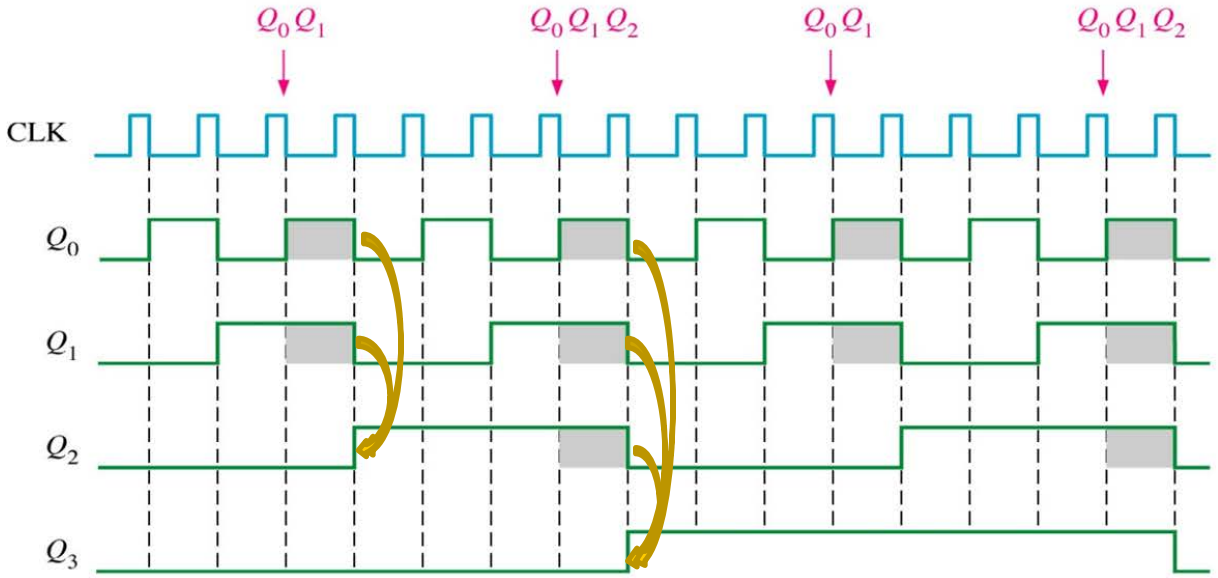


CLOCK PULSE	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8 (recycles)	0	0	0

4-bit synchronous binary counter

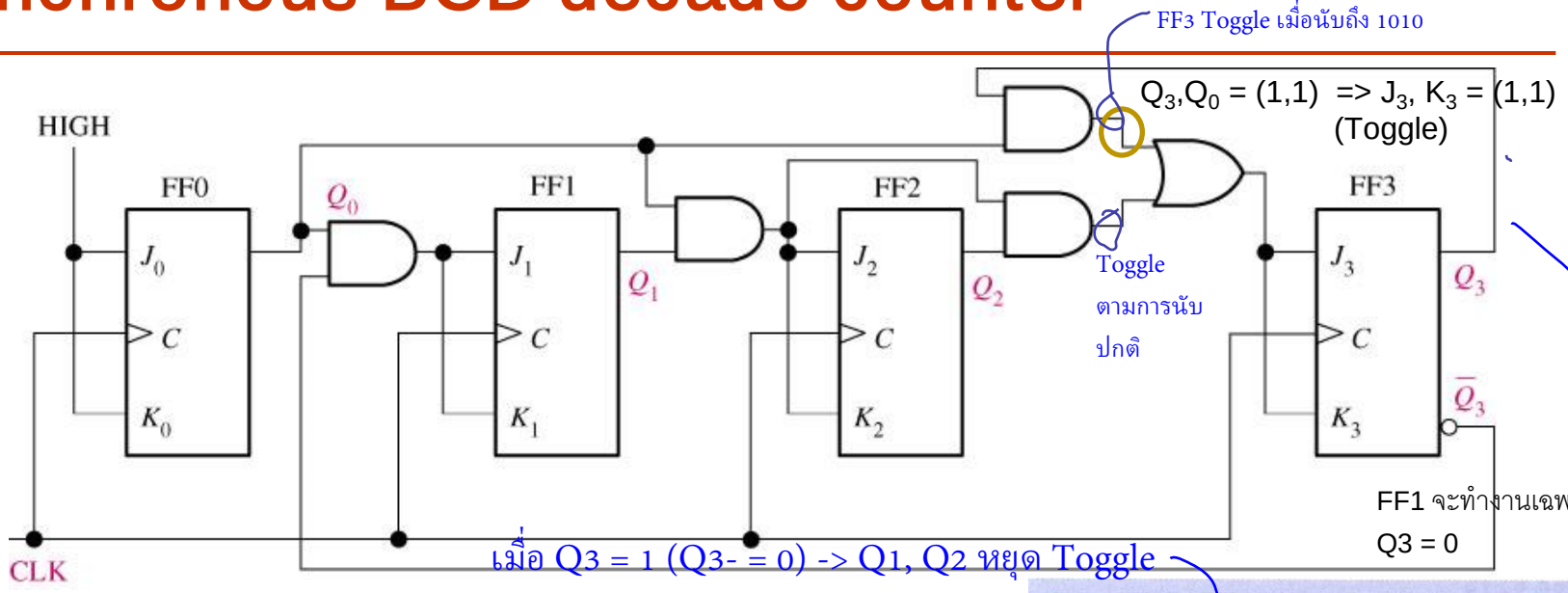


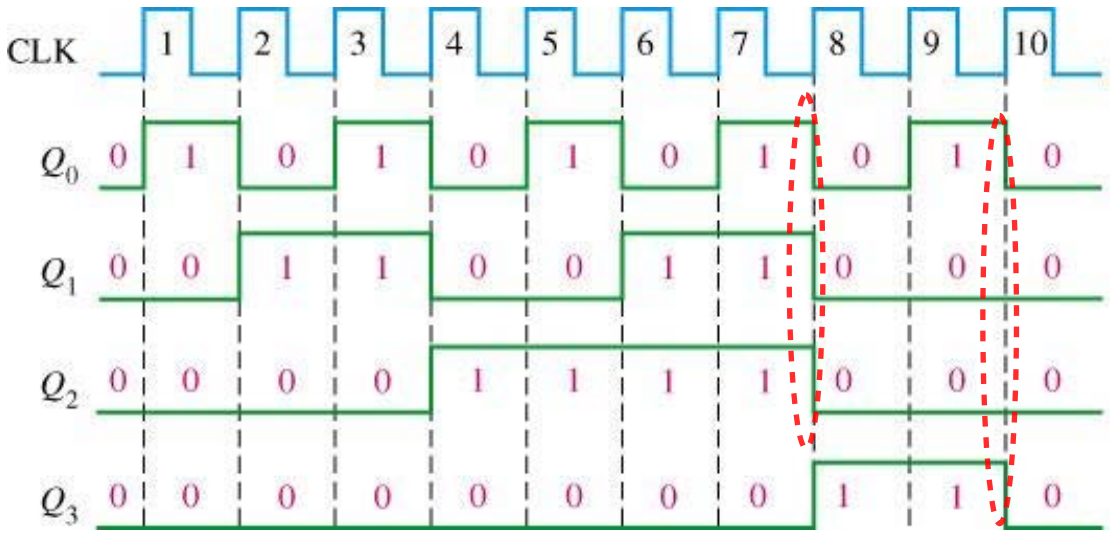
(a)



(b)

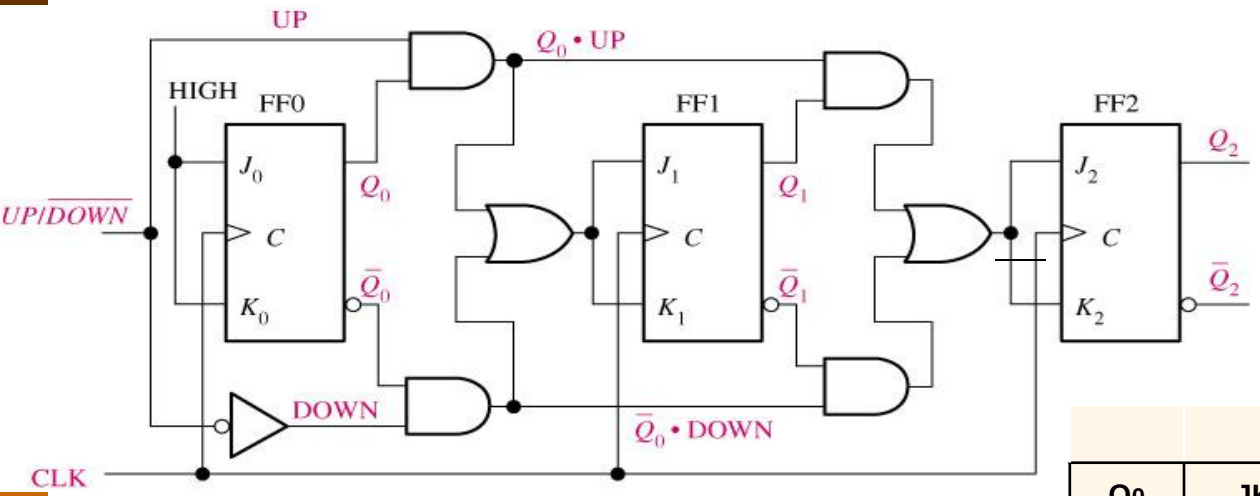
Synchronous BCD decade counter





CLOCK PULSE	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10 (recycles)	0	0	0	0

3-bit up/down synchronous counter

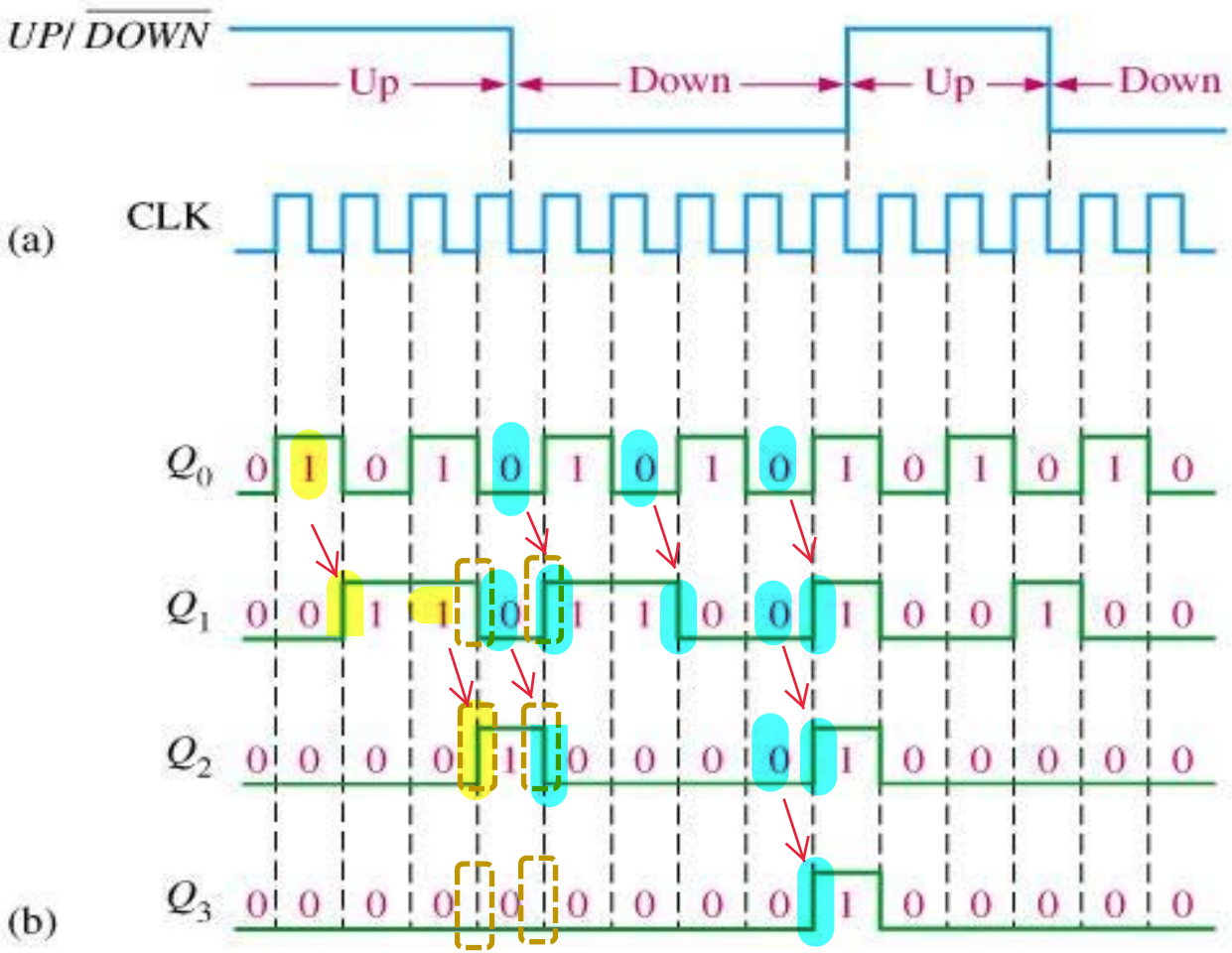


-การทำงานในโหมดนับขึ้นและลง เป็นการเปลี่ยนการทำงานของ JK FF

CLOCK PULSE	UP	Q_2	Q_1	Q_0	DOWN
0		0	0	0	
1		0	0	1	
2		0	1	0	
3		0	1	1	
4		1	0	0	
5		1	0	1	
6		1	1	0	
7		1	1	1	

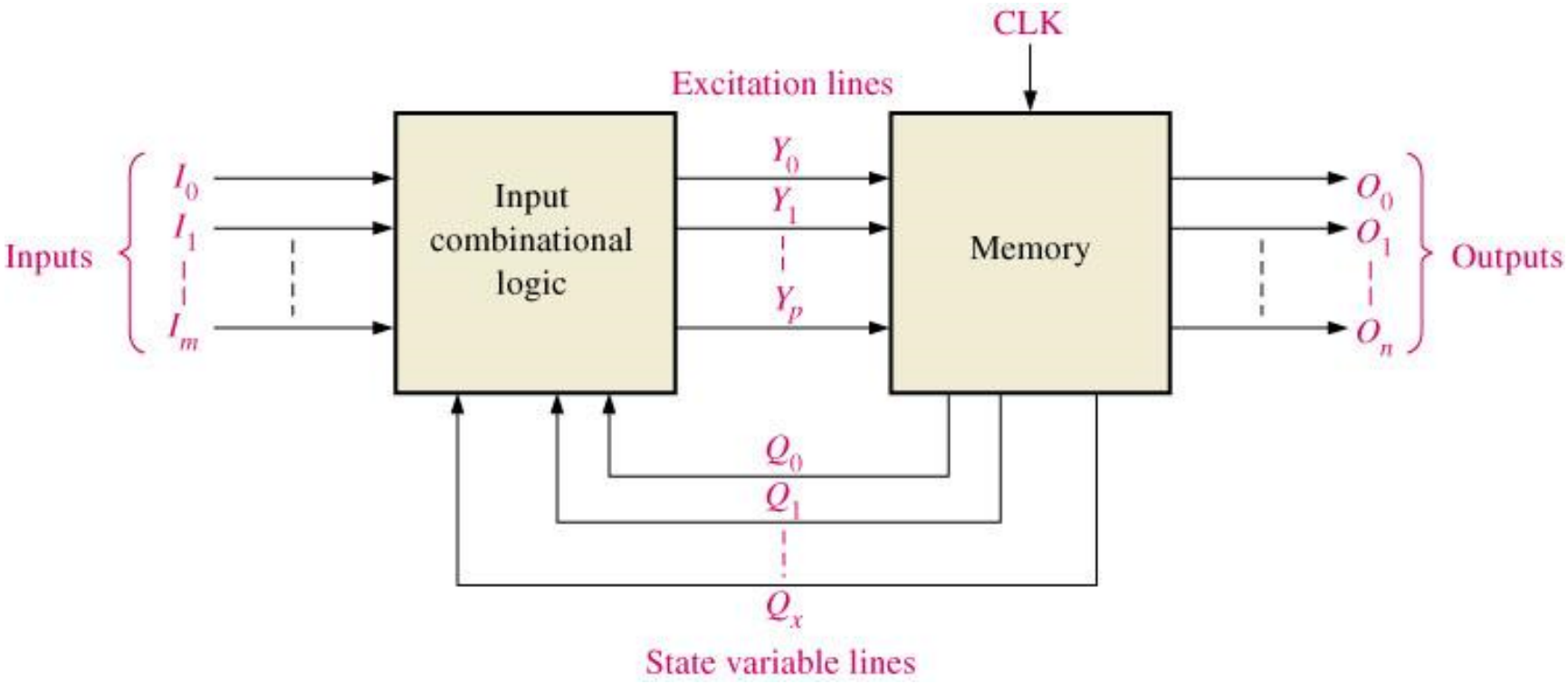
	นับขึ้น			นับลง	
Q_0	JK-FF	Q_1	$\overline{Q}_0$	JK-FF	Q_1
0	no change	0	1	toggle	0
1	toggle	0	0	no change	0
0	no change	1	1	toggle	1
1	toggle	1	0	no change	1
0	no change	0	1	toggle	0
1	toggle	0	0	no change	0
0	no change	1	1	toggle	1
1	toggle	1	0	no change	1

Timing Diagram



Q_3	Q_2	Q_1	Q_0	
0	0	0	0	UP
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	0	1	1	DOWN
0	0	1	0	
0	0	0	1	
0	0	0	0	
1	1	1	1	
0	0	0	0	UP
0	0	0	1	
0	0	1	0	
0	0	0	1	DOWN
0	0	0	0	

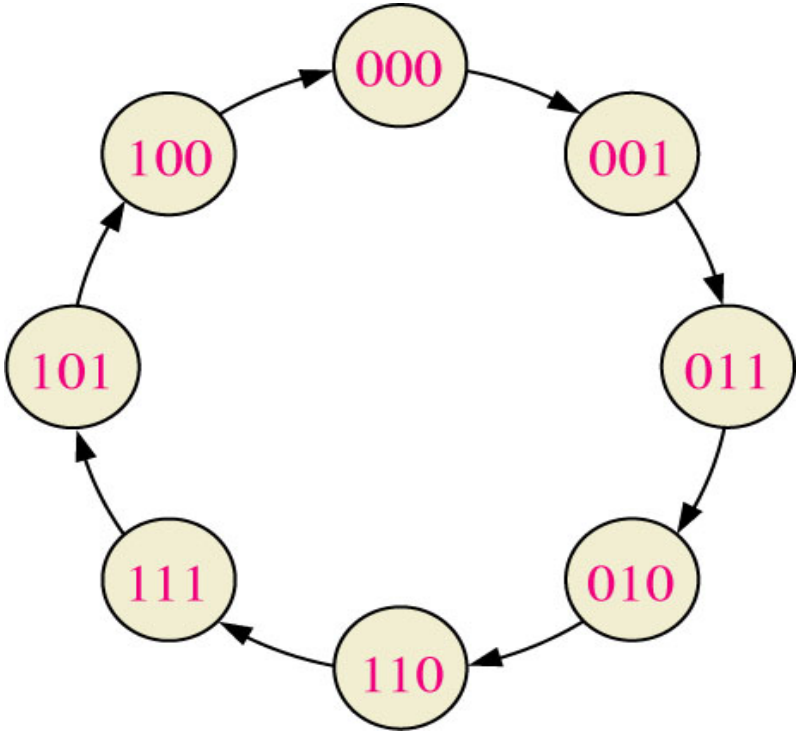
Design of Synchronous Counters



State diagram for a 3-bit Gray code counter

ตัวอย่าง1 จงออกแบบวงจรนับ synchronous แบบ 3-bit Gray code ตาม state diagram ดังรูป **โดยใช้ J-K FF**

Gray code เป็นรหัสเลขฐานสองซึ่งถูกจัดเรียงลำดับให้รหัสข้างเคียงมีบิตเปลี่ยนแปลงได้เพียง 1 บิต

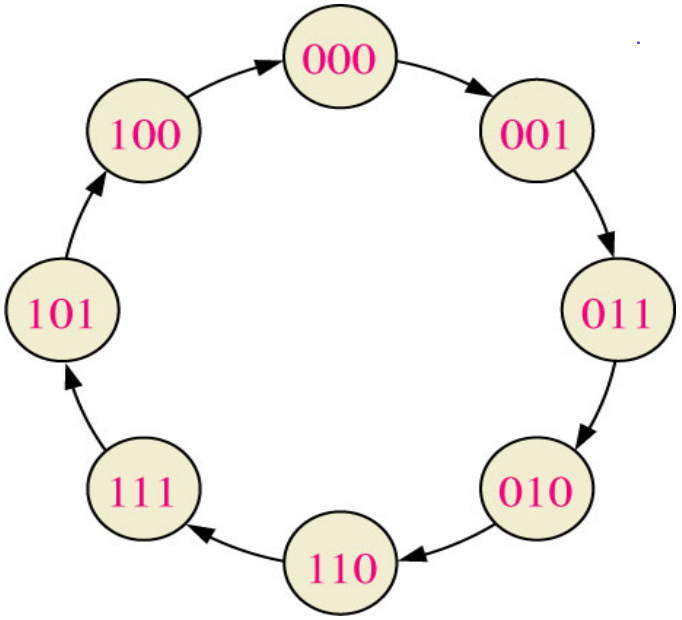


Step 2: Next-Sate Table

เขียน State table จาก State diagram ที่กำหนด

TABLE 9–7

Next-state table for 3-bit Gray code counter.



PRESENT STATE			NEXT STATE		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	0	0	0	1
0	0	1	0	1	1
0	1	1	0	1	0
0	1	0	1	1	0
1	1	0	1	1	1
1	1	1	1	0	1
1	0	1	1	0	0
1	0	0	0	0	0

Step 3: J-K Flip-Flop Transition Table

จาก Transition table (State table) ของ J-K FF

TABLE 9–8

Transition table for a J-K flip-flop.

OUTPUT TRANSITIONS			FLIP-FLOP INPUTS	
Q_N		Q_{N+1}	J	K
0	⟶	0	0	X
0	⟶	1	1	X
1	⟶	0	X	1
1	⟶	1	X	0

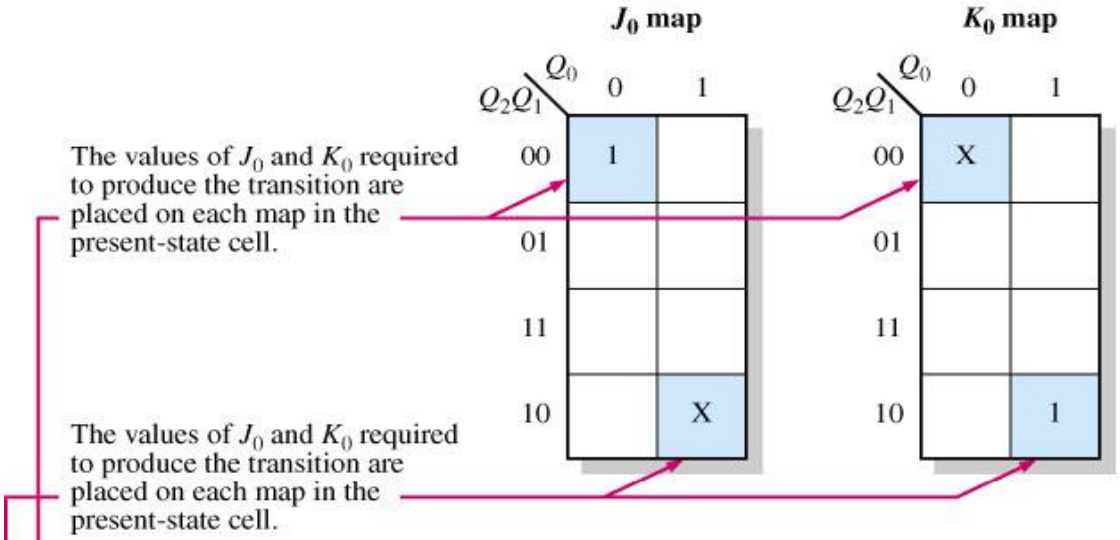
Q_N: present state

Q_{N+1}: next state

X: "don't care"

Step 4: Karnaugh Maps

ทำการหา Excite table จาก state table ของวงจรและ J-K FF



Output Transitions Q_N	Q_{N+1}	Flip-Flop Inputs J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Flip-flop transition table

Present State Q_2	Q_1	Q_0	Next State Q_2	Q_1	Q_0
0	0	0	0	0	1
0	0	1	0	1	1
0	1	1	0	1	0
0	1	0	1	1	0
1	1	0	1	1	1
1	1	1	1	0	1
1	0	1	1	0	0
1	0	0	0	0	0

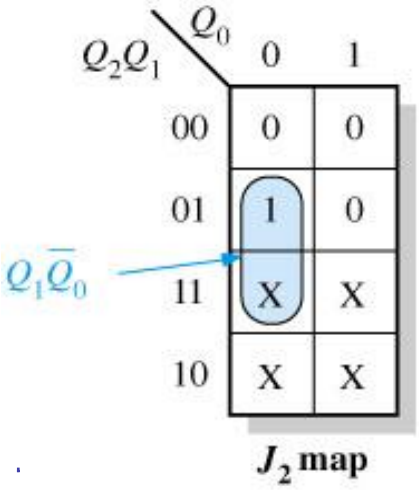
Next-state table

For the present state 000, Q_0 makes a transition from 0 to 1 to the next state.

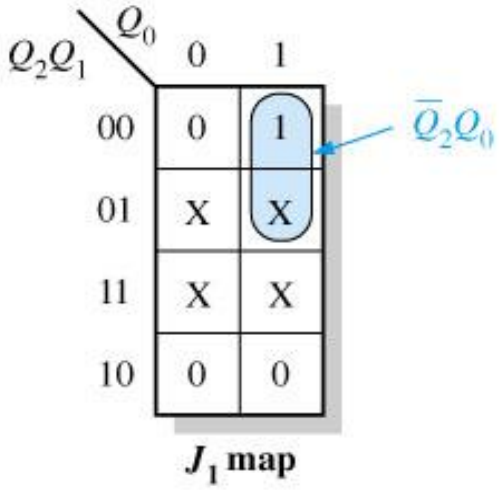
For the present state 101, Q_0 makes a transition from 1 to 0 to the next state.

Karnaugh maps for present-state J and K inputs

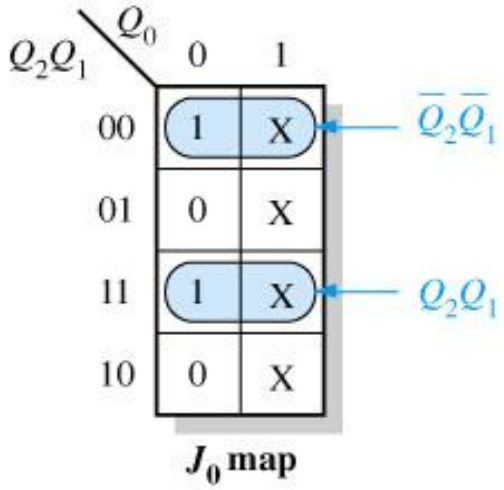
ทำการลดรูปโดย KM และเขียน logic function ของอินพุต JK-FF



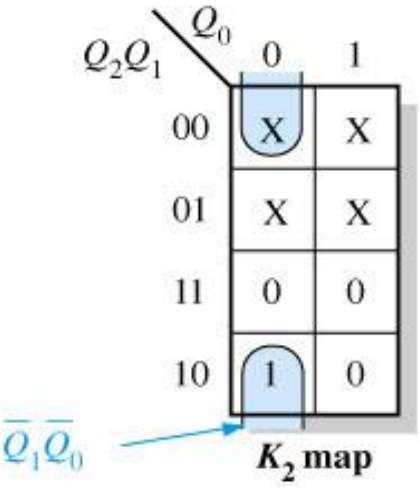
J_2 map



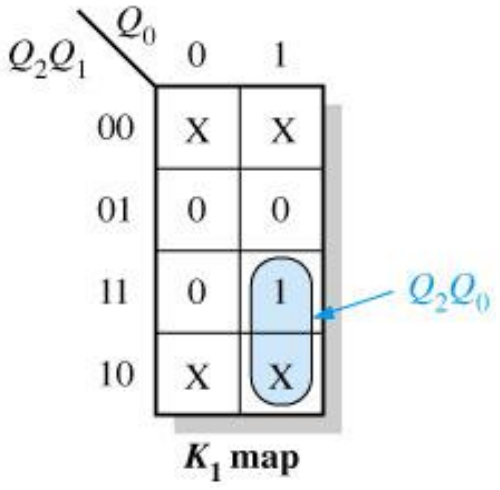
J_1 map



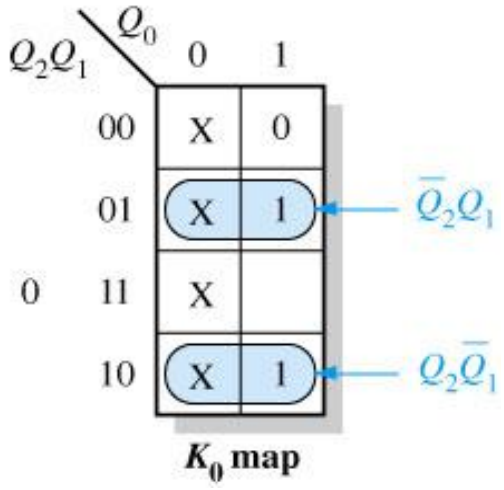
J_0 map



K_2 map



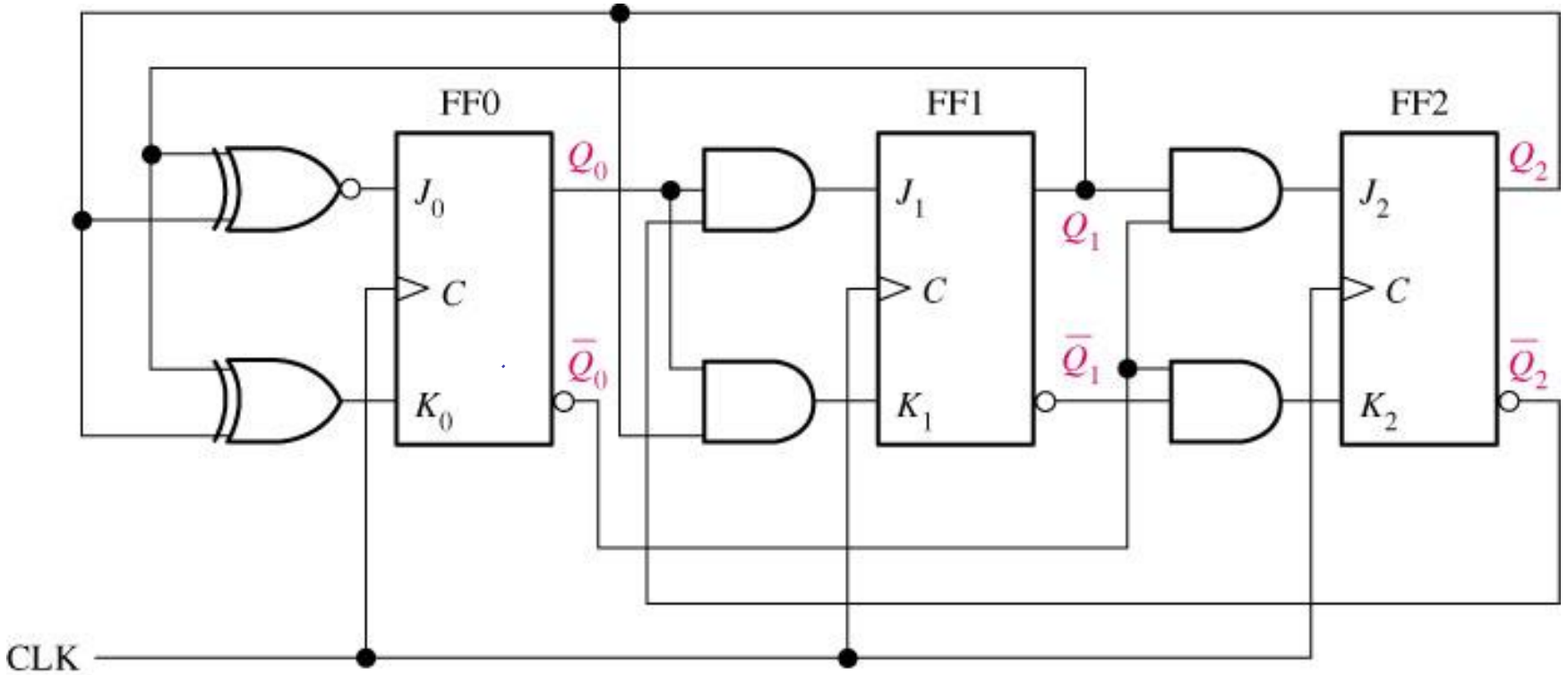
K_1 map



K_0 map

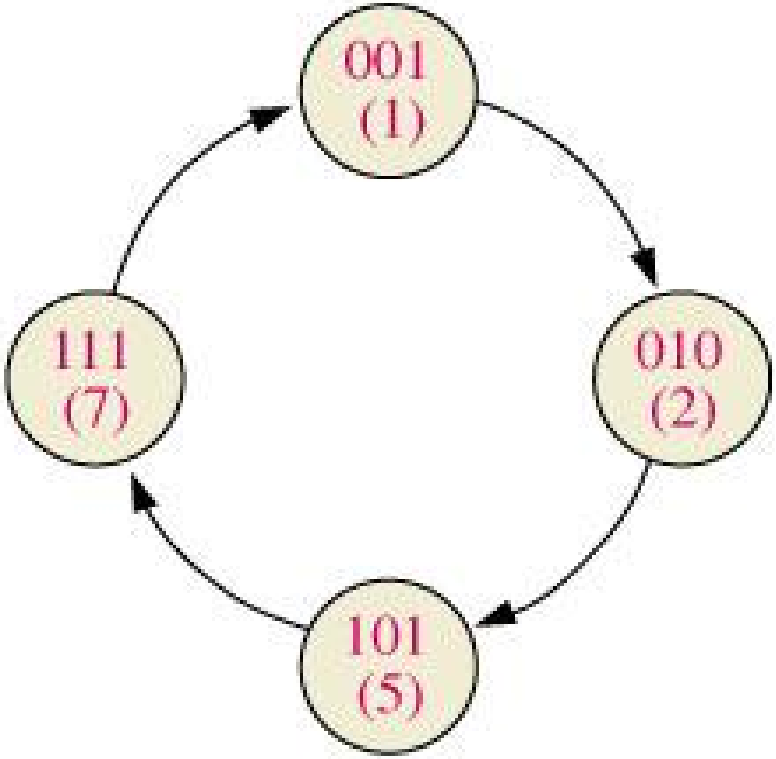
Step 5: 3-bit Gray code counter

ทำการสังเคราะห์วงจรจาก logic function ของอินพุต JK-FF



State Diagram for a 1-2-5-7 Counter

ตัวอย่าง2 จงออกแบบวงจรนับ synchronous แบบ 3-bit ตาม state diagram ดังรูป **โดยใช้ J-K FF**



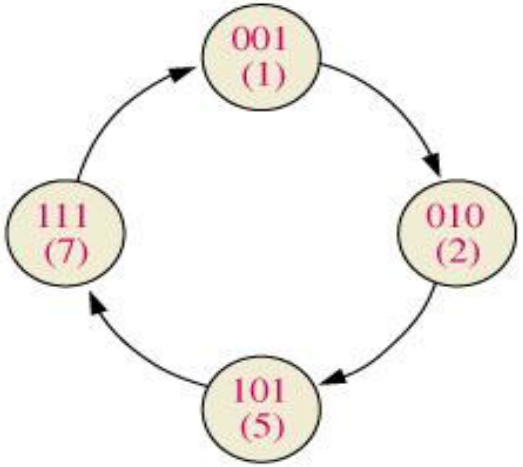
Step 2: Next-Sate Table

เขียน State table จาก State diagram ที่กำหนด

TABLE 9–9

Next-state table.

PRESENT STATE			NEXT STATE		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	1	0	1	0
0	1	0	1	0	1
1	0	1	1	1	1
1	1	1	0	0	1



Step 3: J-K Flip-Flop Transition Table

จาก Transition table (State table) ของ J-K FF

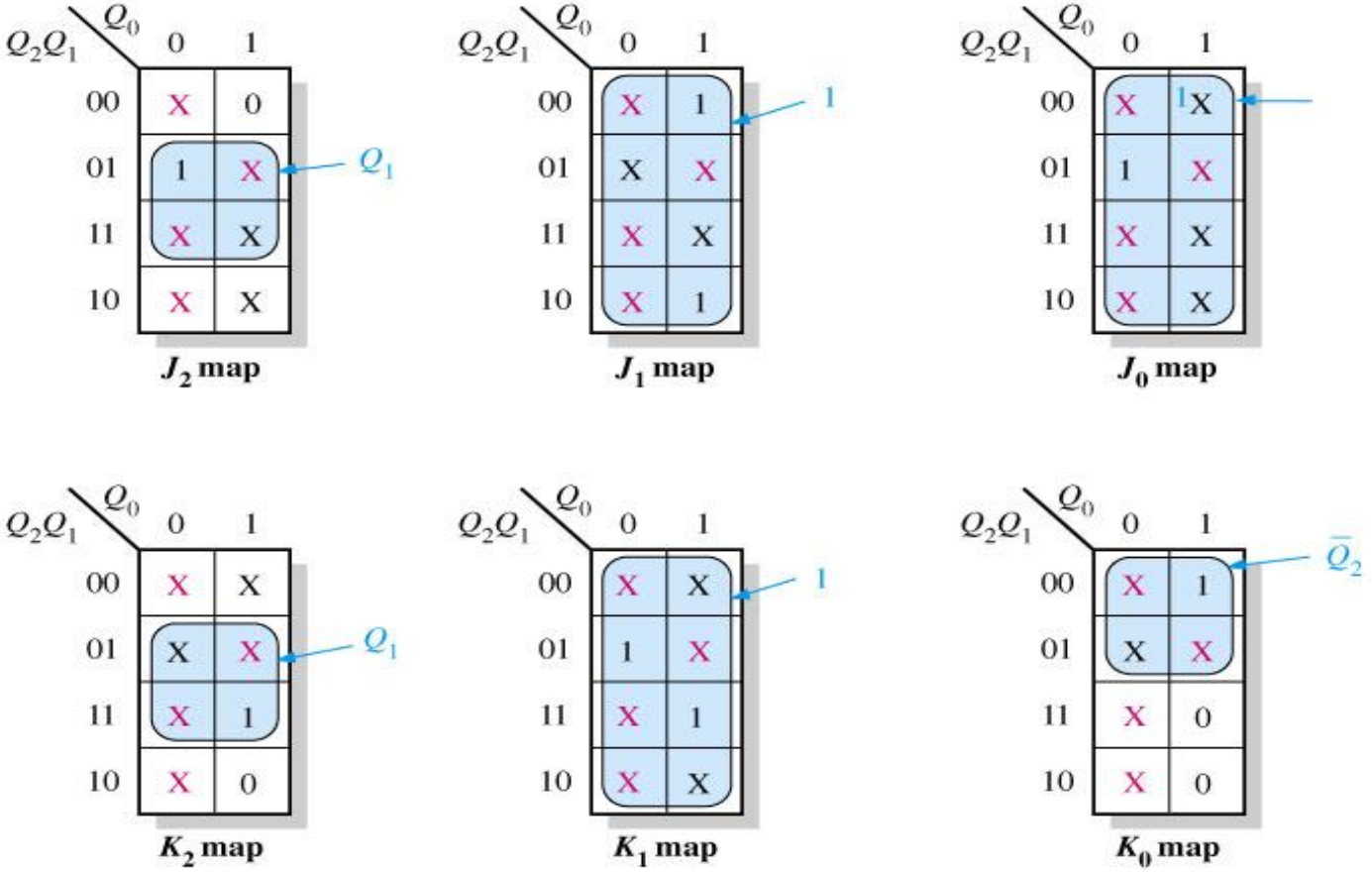
TABLE 9–10

Transition table for a J-K flip-flop.

OUTPUT TRANSITIONS			FLIP-FLOP INPUTS	
Q_N		Q_{N+1}	J	K
0	⟶	0	0	X
0	⟶	1	1	X
1	⟶	0	X	1
1	⟶	1	X	0

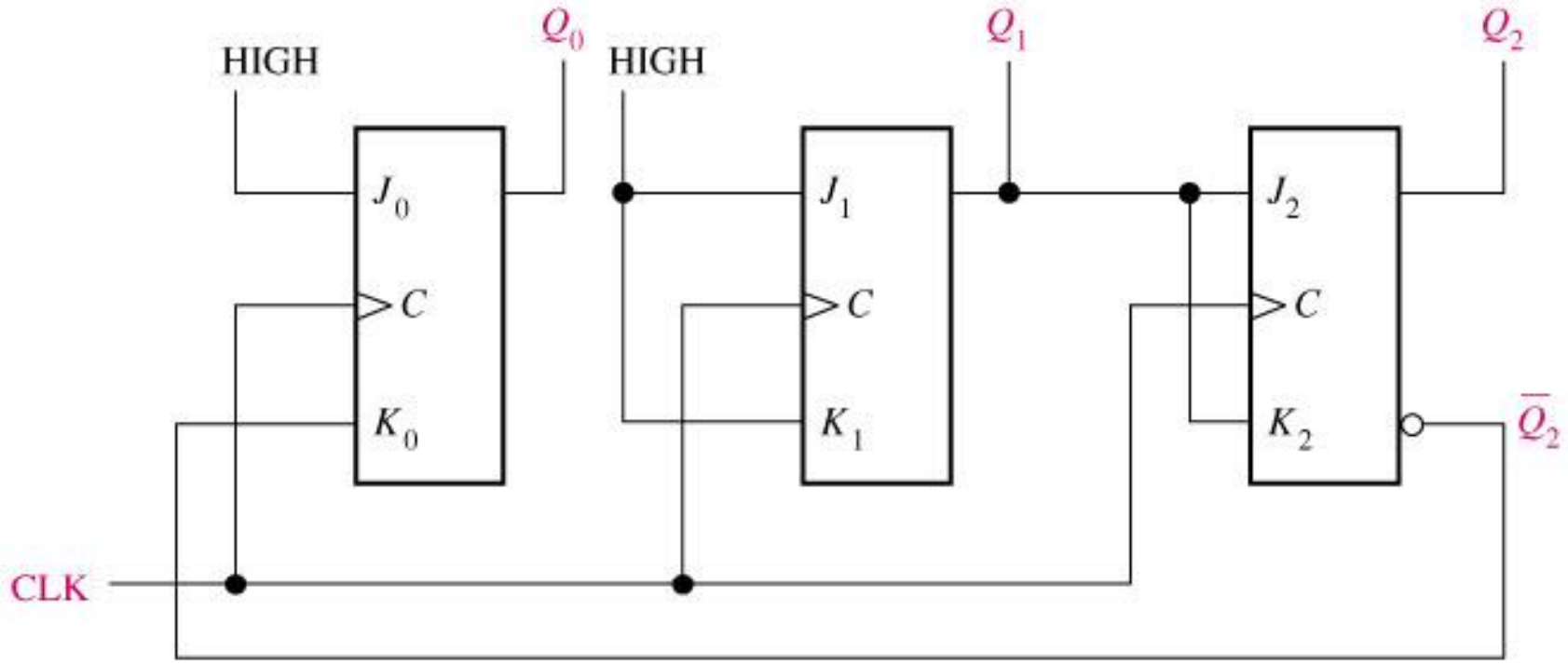
Step 4: Karnaugh Maps

ทำการหา Excite table จาก state table ของวงจรและ J-K FF
ทำการลดรูปโดย KM และเขียน logic function ของอินพุต JK-FF



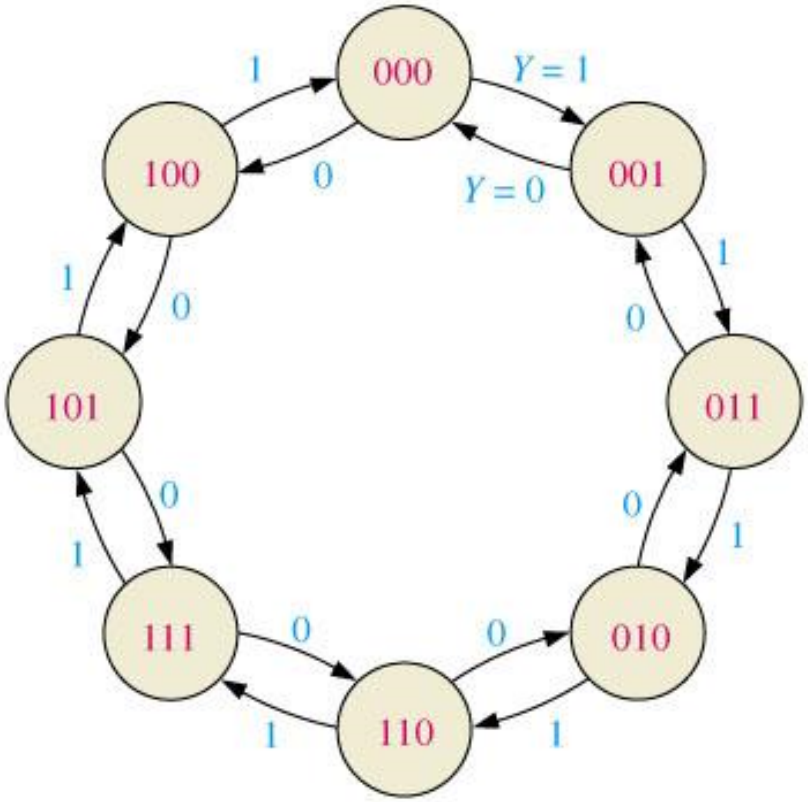
Step 5: Implementation of 1-2-5-7 Counter

ทำการสังเคราะห์วงจรจาก logic function ของอินพุต JK-FF



State diagram for a 3-bit up/down Gray code counter

ตัวอย่าง3 จงออกแบบวงจรนับ synchronous แบบ 3-bit ตาม state diagram ดังรูป **โดยใช้ J-K FF**



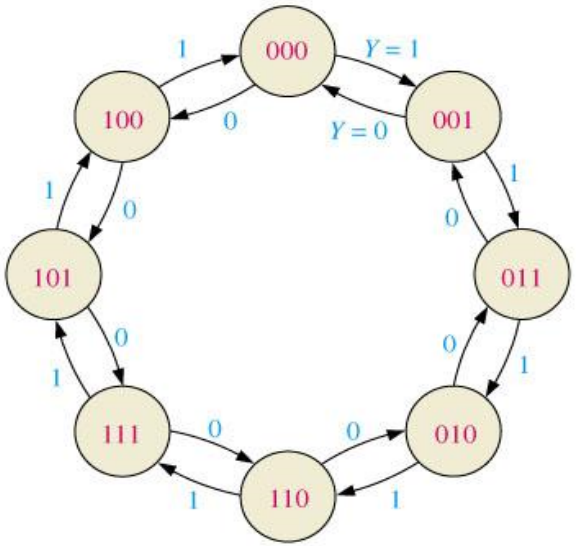
Step 2: Next-Sate Table

TABLE 9–11

Next-state table for 3-bit up/down Gray code counter.

PRESENT STATE			NEXT STATE Y = 0 (DOWN)			NEXT STATE Y = 1 (UP)		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	0	1	0	0	0	0	1
0	0	1	0	0	0	0	1	1
0	1	1	0	0	1	0	1	0
0	1	0	0	1	1	1	1	0
1	1	0	0	1	0	1	1	1
1	1	1	1	1	0	1	0	1
1	0	1	1	1	1	1	0	0
1	0	0	1	0	1	0	0	0

Y = UP/$\overline{\text{DOWN}}$ control input.



Step 3: Flip-Flop Transition Table

จาก Transition table (State table) ของ J-K FF

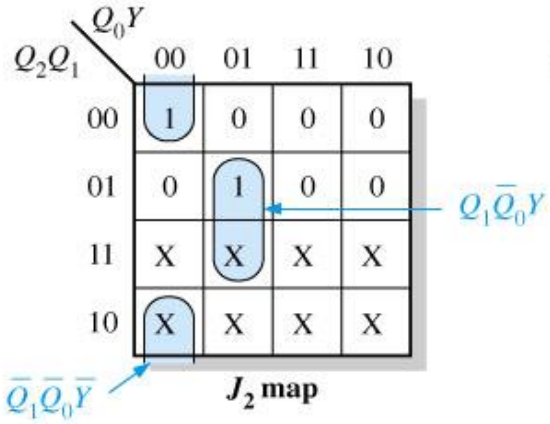
TABLE 9–12

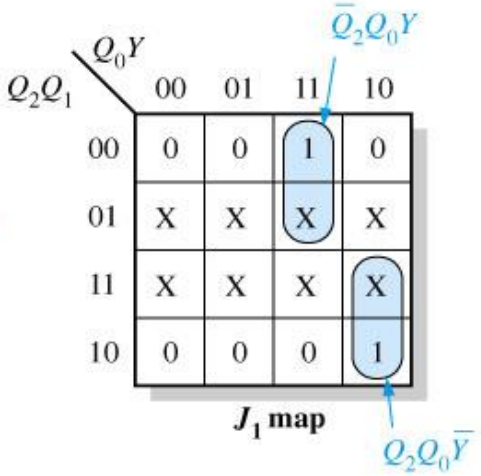
Transition table for a J-K flip-flop.

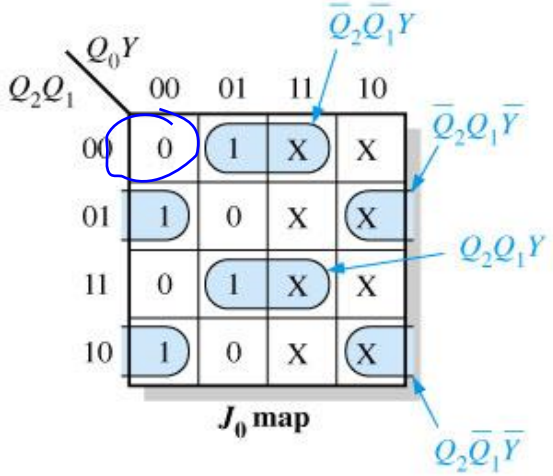
OUTPUT TRANSITIONS			FLIP-FLOP INPUTS	
Q_N		Q_{N+1}	J	K
0	⟶	0	0	X
0	⟶	1	1	X
1	⟶	0	X	1
1	⟶	1	X	0

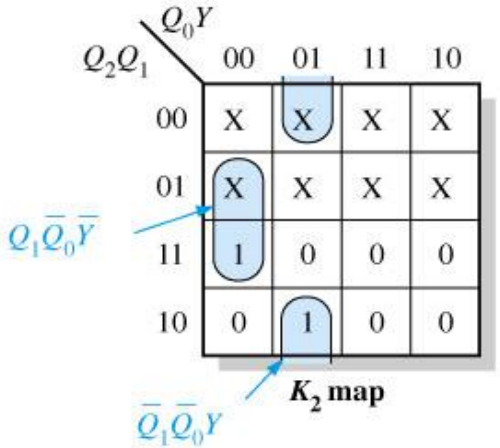
J and K maps. Y is UP/DOWN control input

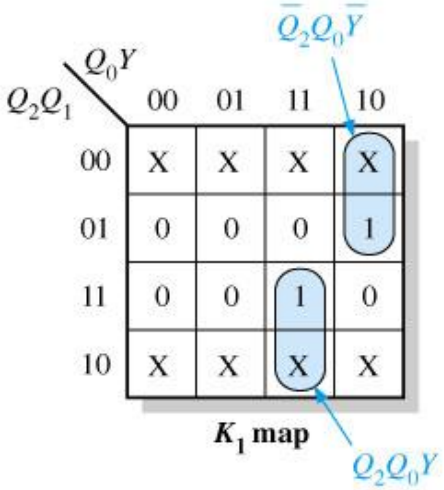
ทำการหา Excite table จาก state table ของวงจรและ J-K FF
ทำการลดรปโดย KM และเขียน logic function ของอินพต JK-FF

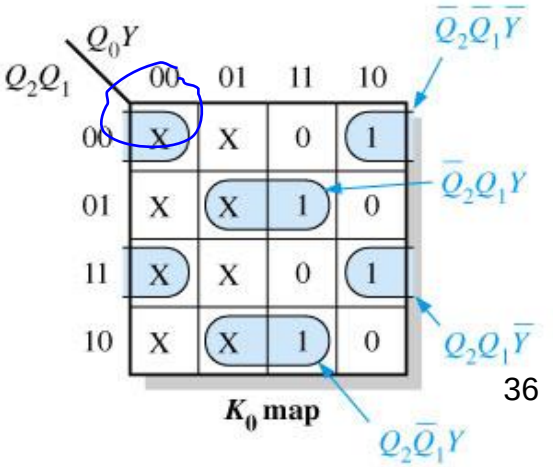






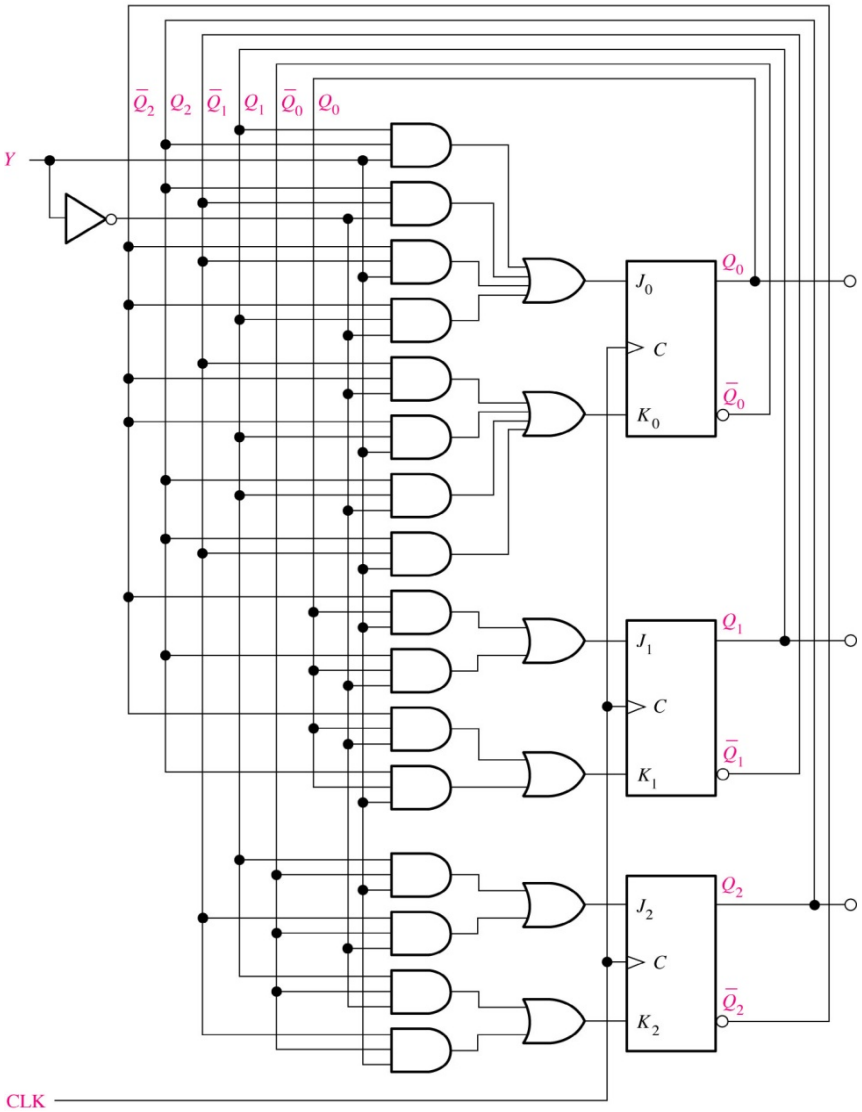






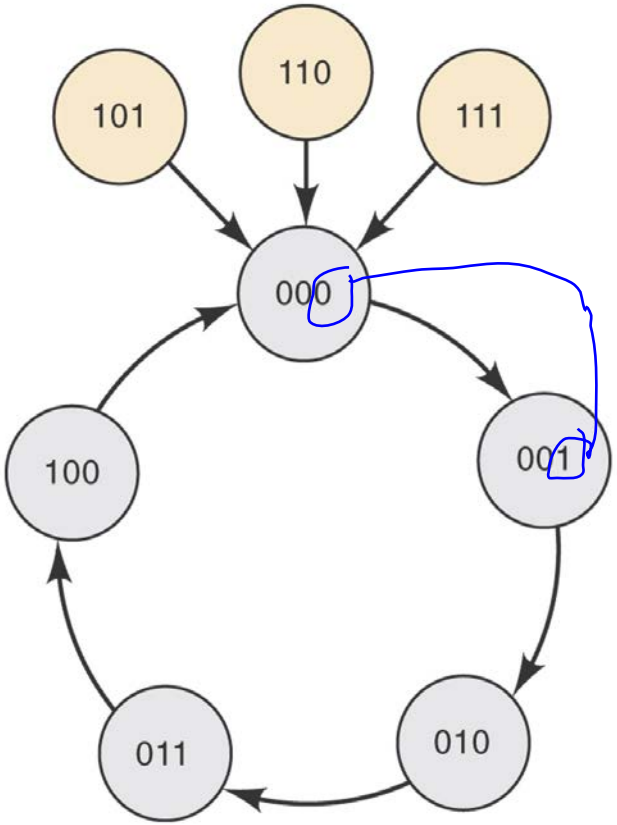
3-bit up/down Gray code counter

ทำการสังเคราะห์วงจรจาก logic function ของอินพุต JK-FF



State transition diagram for the synchronous counter design example

<u>การบ้าน 1</u> จงออกแบบวงจรนับ synchronous แบบ 3-bit ตาม state diagram ดังรูป **<u>โดยใช้ J-K FF</u>**



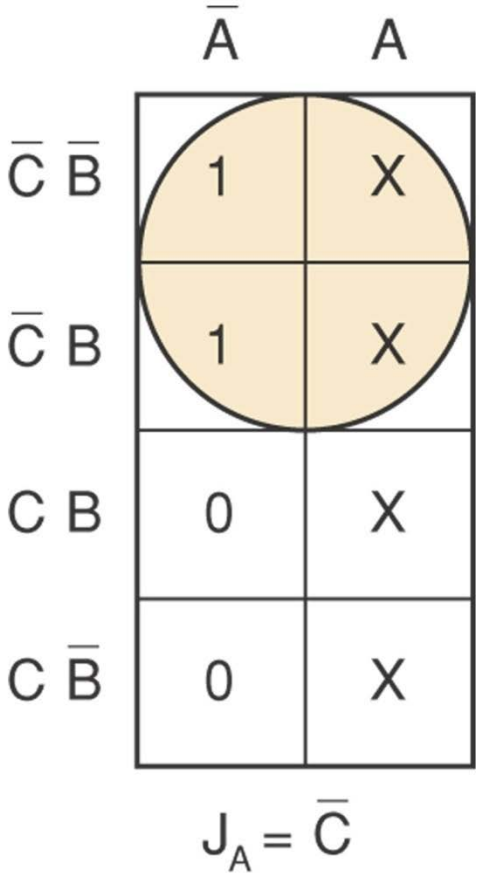
$J_A = 1, K_A = X$

K map used to obtain the simplified expression for J_A

(hint: การบ้าน 1)

PRESENT			
C	B	A	J_A
0	0	0	1
0	0	1	x
0	1	0	1
0	1	1	x
1	0	0	0
1	0	1	x
1	1	0	0
1	1	1	x

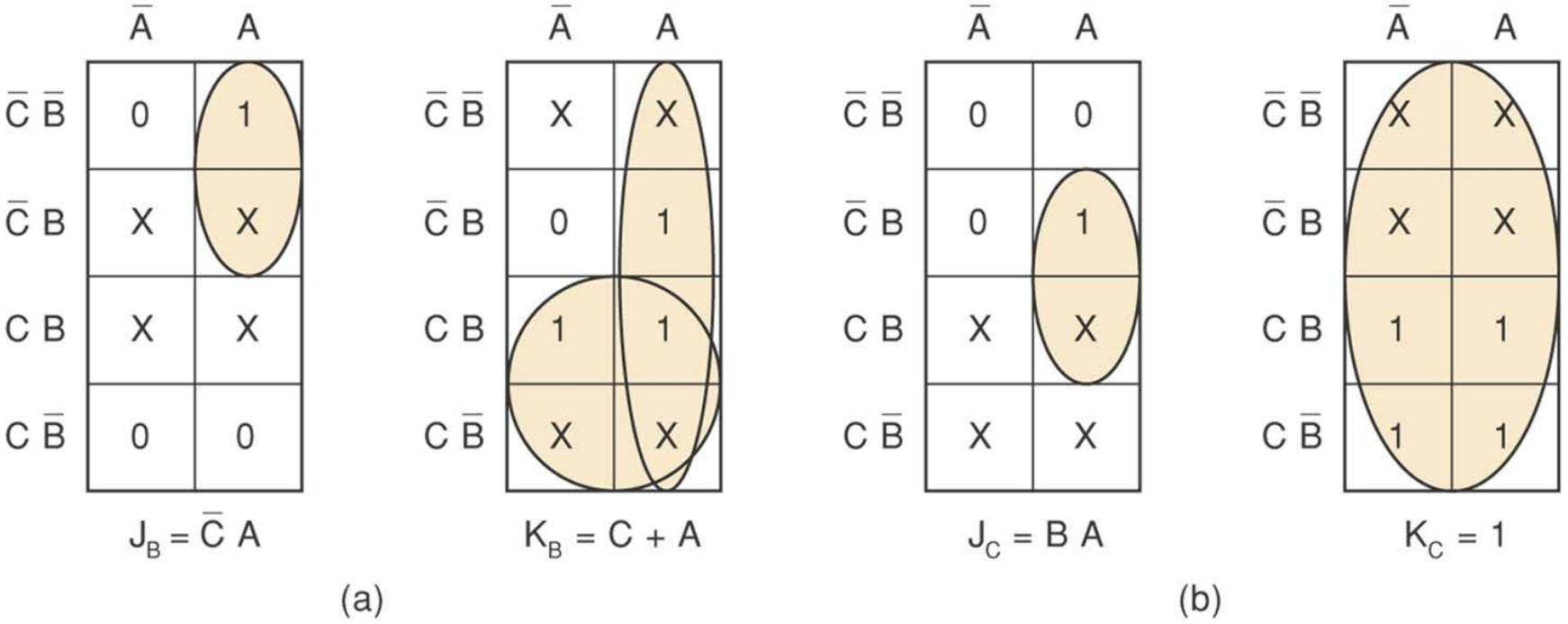
(a)



(b)

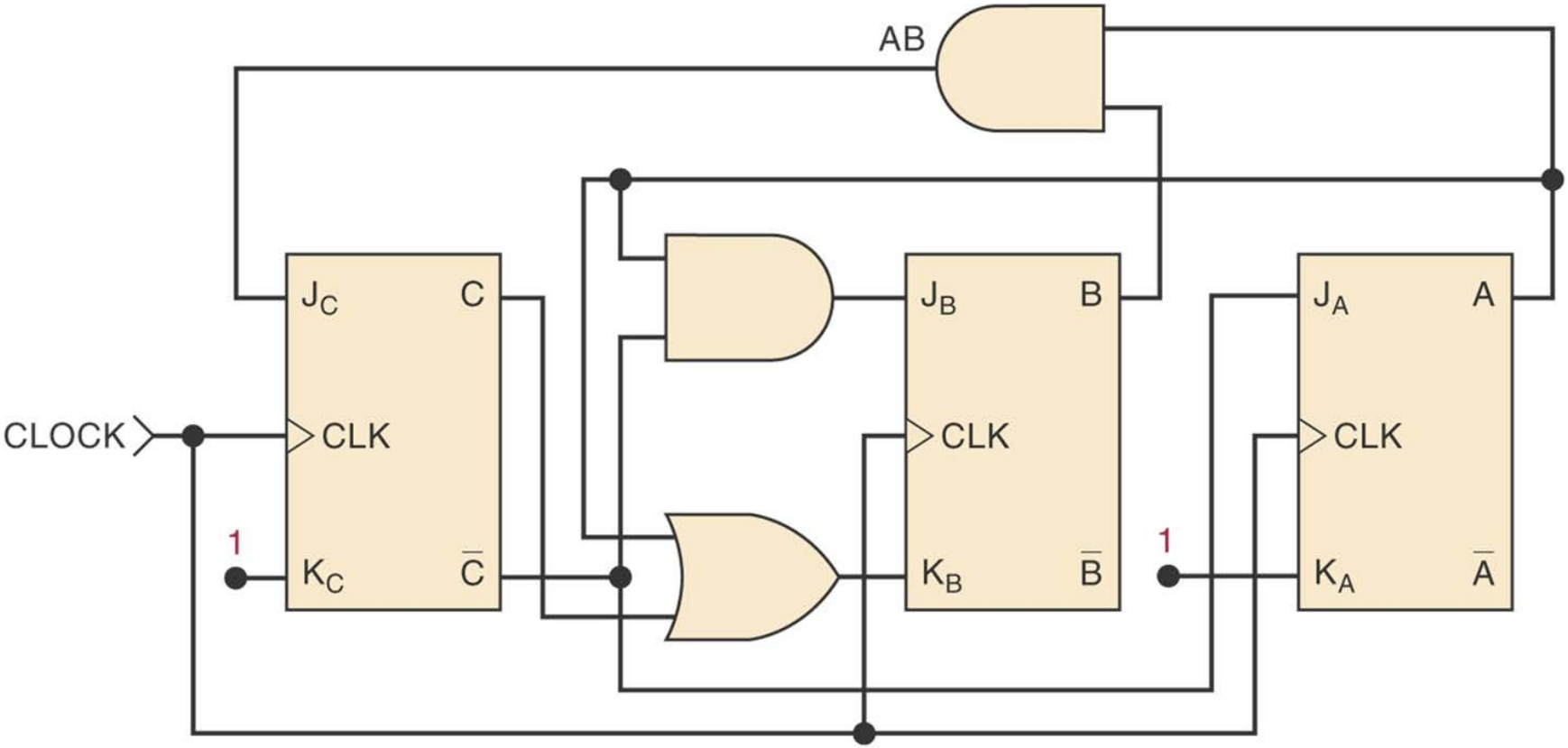
K-maps for the J_B, K_B, J_C and K_C logic circuits

(hint: การบ้าน 1)



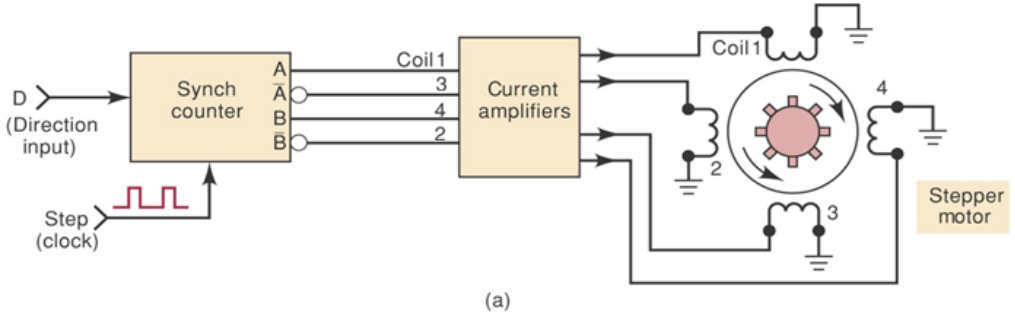
Implementation of the synchronous counter

คำตอบ การบ้าน1

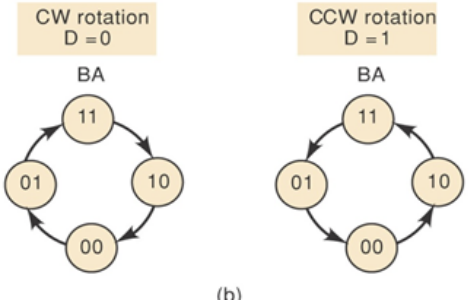


(a) synchronous supplies the sequential outputs to drive a stepper motor
(b) state transition diagrams for both states of Direction input, D

<u>การบ้าน 2</u> จงออกแบบวงจรนับแบบ synchronous ของรหัส Gray แบบ 2-bits **<u>โดยใช้ JK-FF</u>** สำหรับใช้ขับ stepping motor ตาม state diagram ด้านล่าง กำหนด ให้ D เป็นสัญญาณควบคุมการนับเพิ่ม/ลด (ควบคุมทิศทางการหมุนของมอเตอร์)

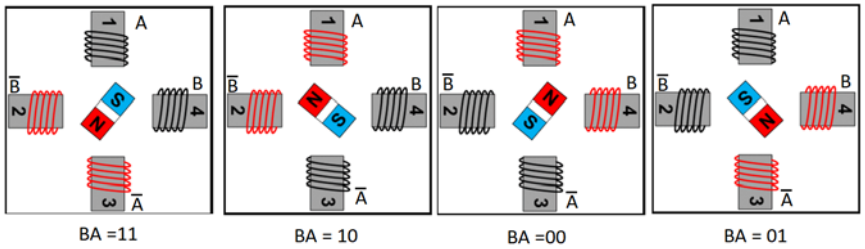


(a)



(b)

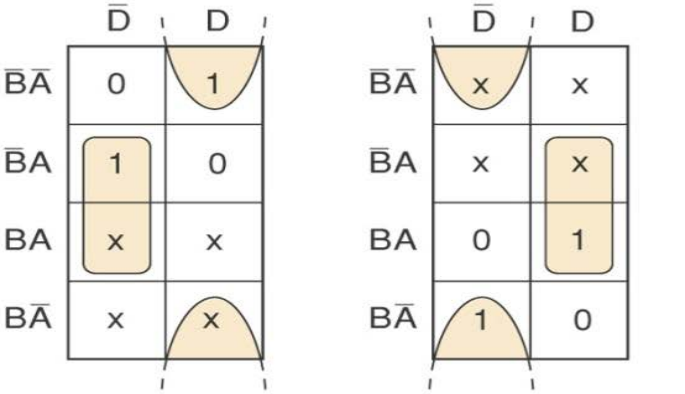
D	BP	AP	BN	AN	JB	KB	JA	KA
0	1	1	1	0	d	0	d	1
0	1	0	0	0	d	1	0	d
0	0	0	0	1	0	d	1	d
0	0	1	1	1	1	d	d	0
1	1	1	0	1	d	1	d	0
1	0	1	0	0	0	d	d	1
1	0	0	1	0	1	d	0	d
1	1	0	1	1	d	0	1	d



Karnaugh maps for A and B FF

(hint: การบ้าน 2)

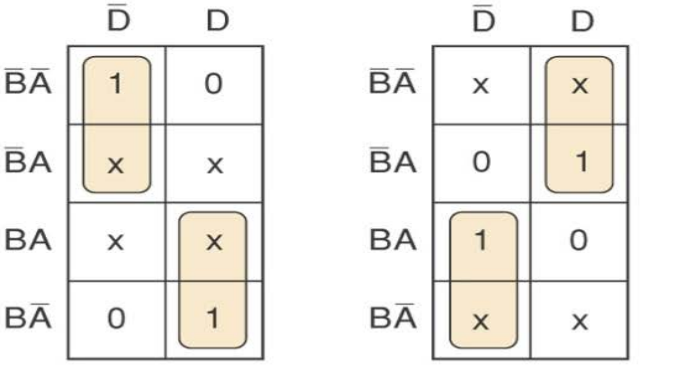
D	BP	AP	BN	AN	JB	KB	JA	KA
0	0	0	0	1	0	d	1	d
0	0	1	1	1	1	d	d	0
0	1	0	0	0	d	1	0	d
0	1	1	1	0	d	0	d	1
1	0	0	1	0	1	d	0	d
1	0	1	0	0	0	d	d	1
1	1	0	1	1	d	0	1	d
1	1	1	0	1	d	1	d	0



$J_B = \bar{D}A + D\bar{A}$
$= D \oplus A$

$K_B = \bar{D}\bar{A} + DA$
$= \overline{D \oplus A}$

(a)



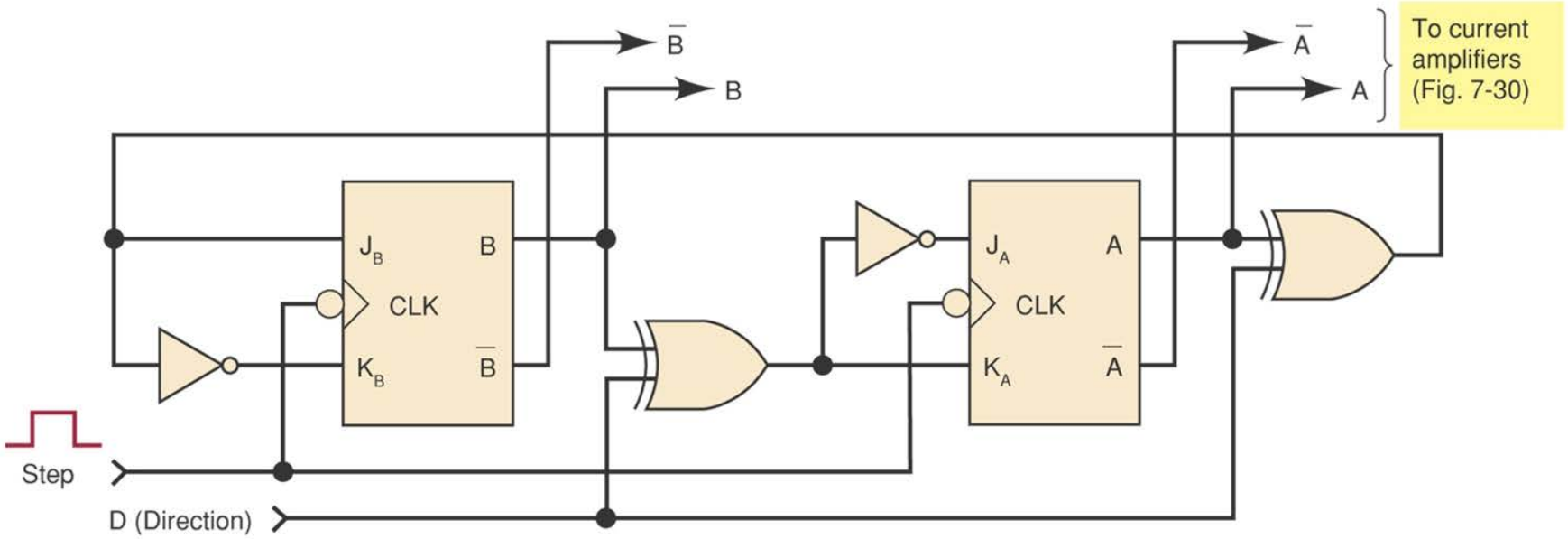
$J_A = \bar{D}\bar{B} + DB$
$= \overline{D \oplus B}$

$K_A = \bar{D}B + D\bar{B}$
$= D \oplus B$

(b)

Synchronous counter implementation

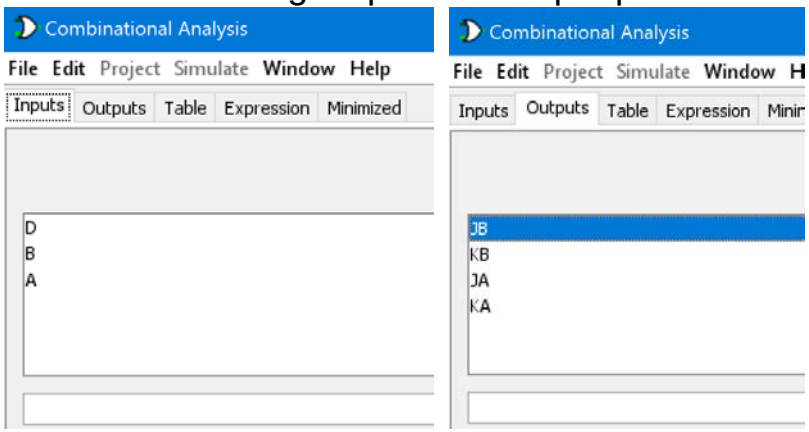
<u>คำตอบ</u> การบ้าน2



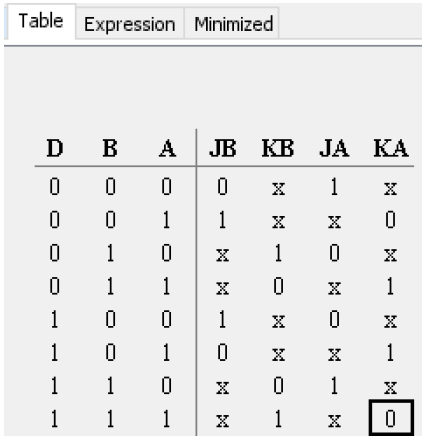
Synchronous counter implementation

คำตอบ การบ้าน2

1.Assign input and output pin

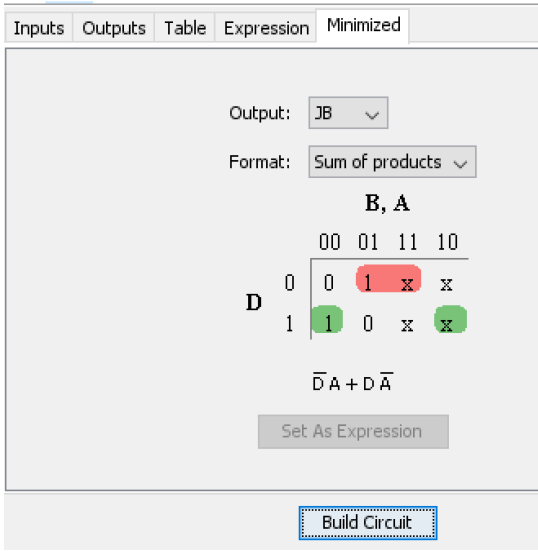


2. Create Excite-table

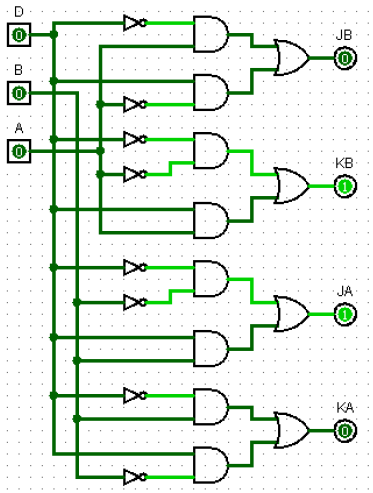


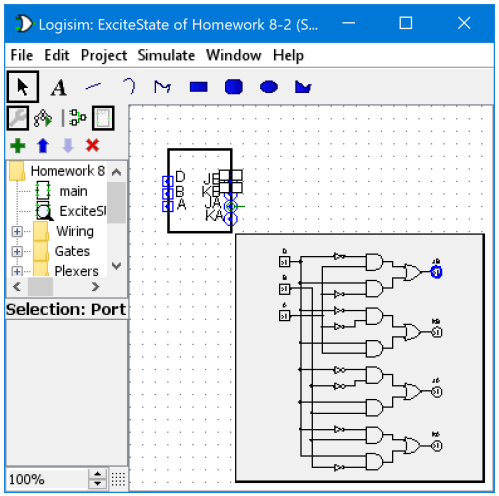
D	B	A	JB	KB	JA	KA
0	0	0	0	x	1	x
0	0	1	1	x	x	0
0	1	0	x	1	0	x
0	1	1	x	0	x	1
1	0	0	1	x	0	x
1	0	1	0	x	x	1
1	1	0	x	0	1	x
1	1	1	x	1	x	0

3. Check expression and build excite-circuit

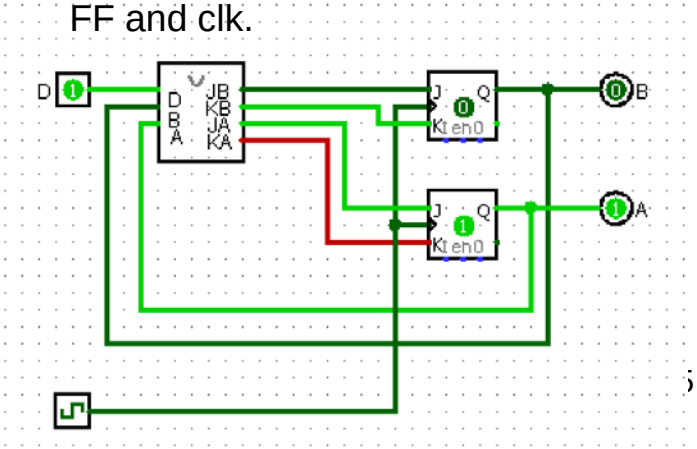


4. Edit excite-circuit appearance

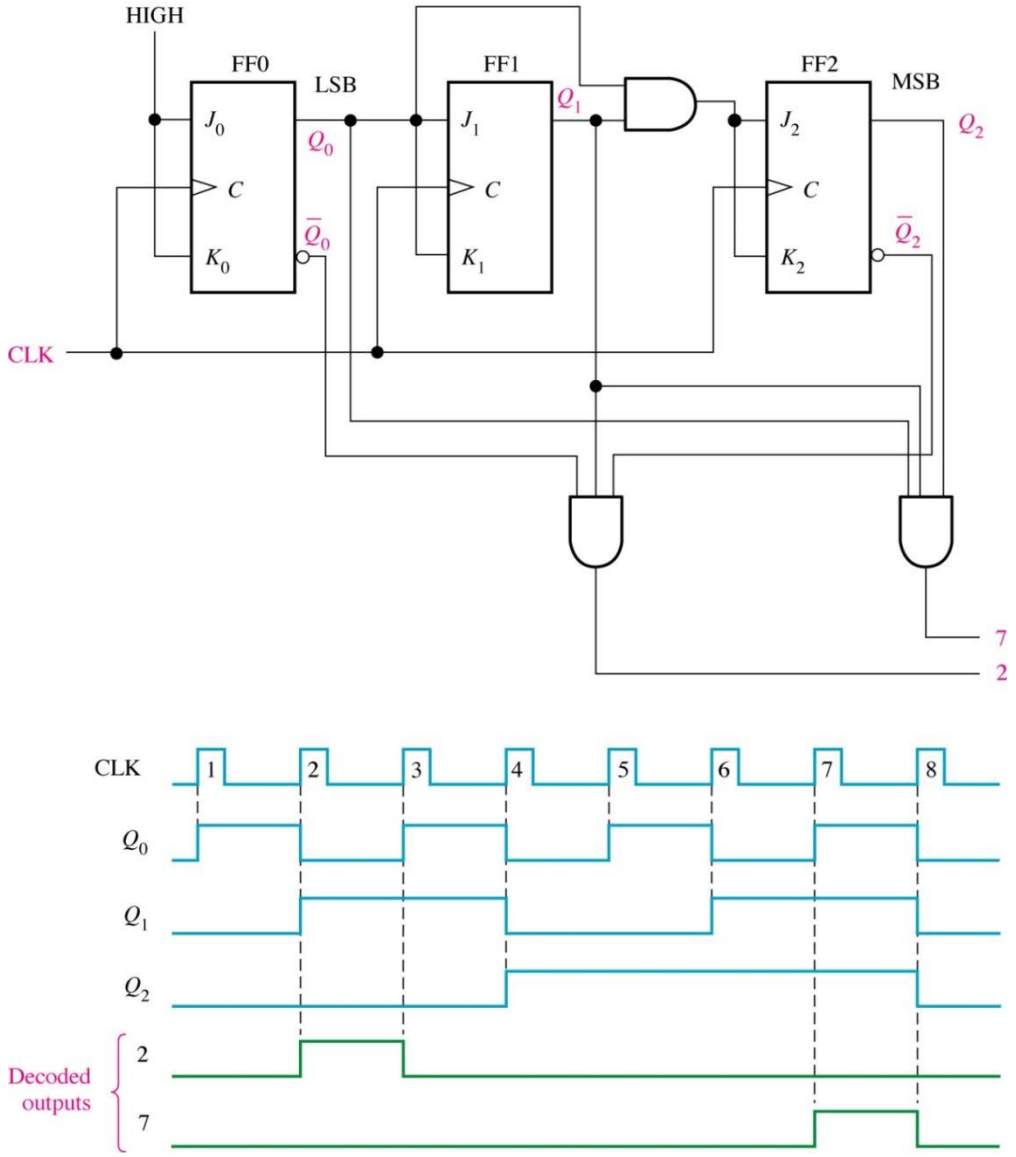




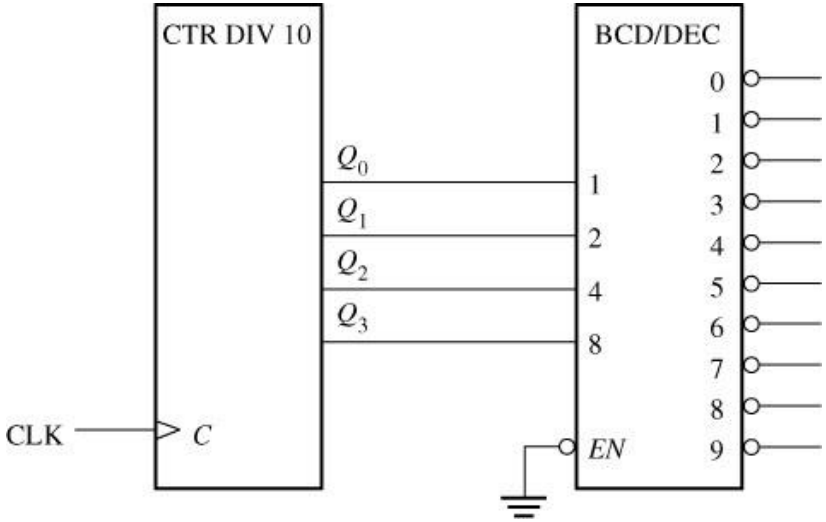
5. Complete whole circuit with excite-circuit, FF and clk.

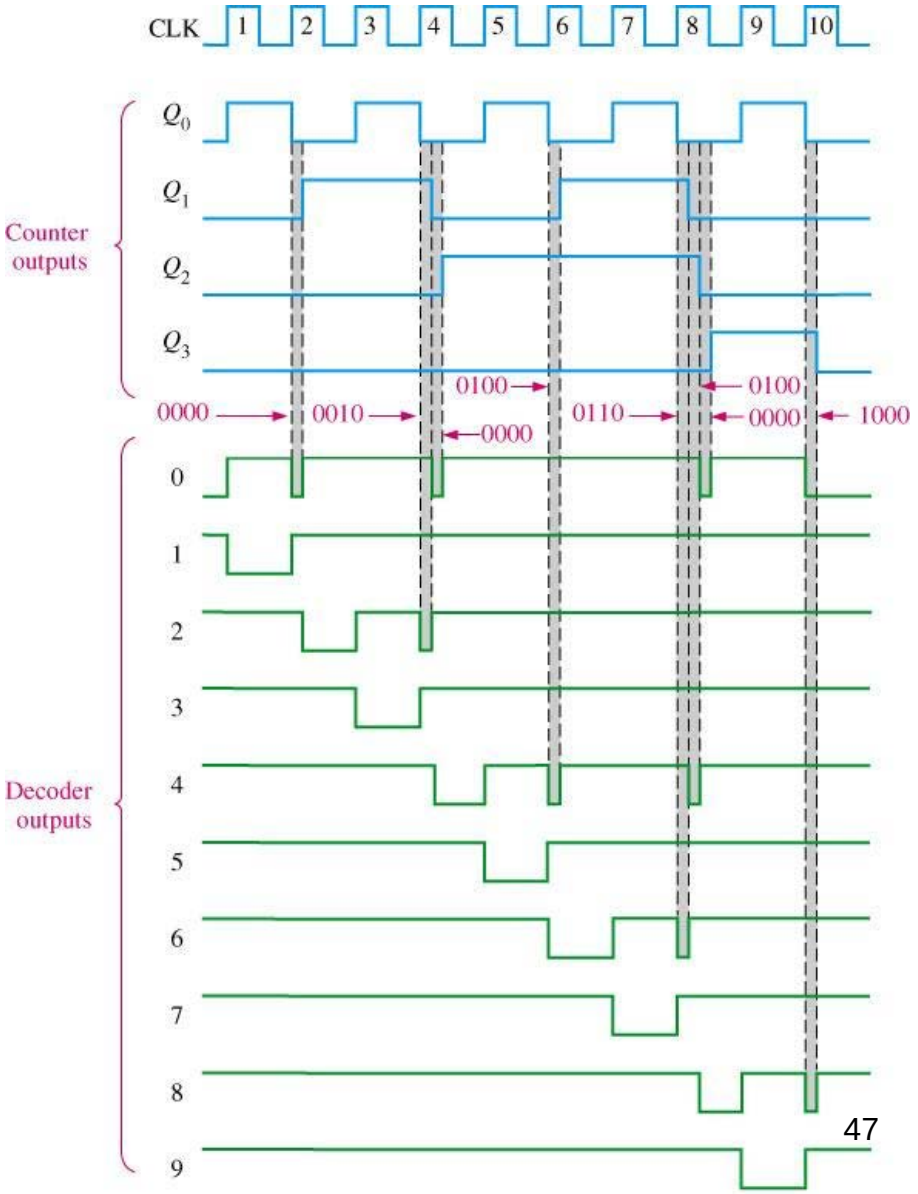


A 3-bit counter with decoding of count 2 and 7



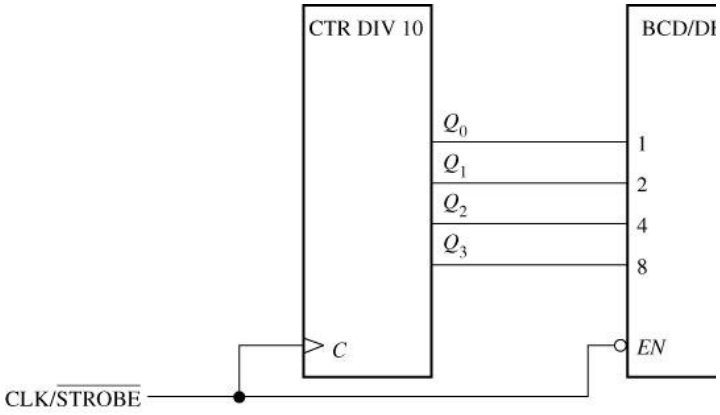
Basic decade (BCD) counter and decoder

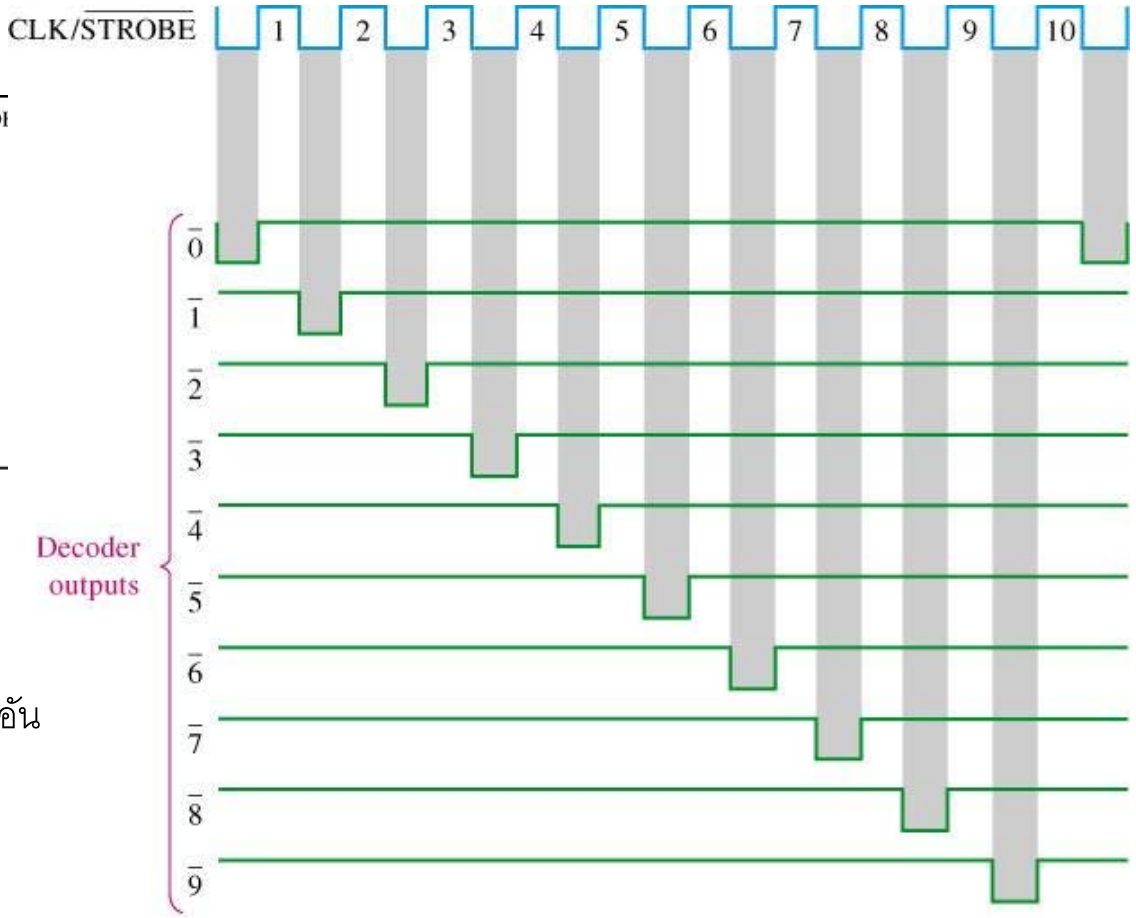




Propagation delay ของ Counter (CTR DIV10) จะทำให้เกิด glitch ขึ้นที่ BCD Decoder

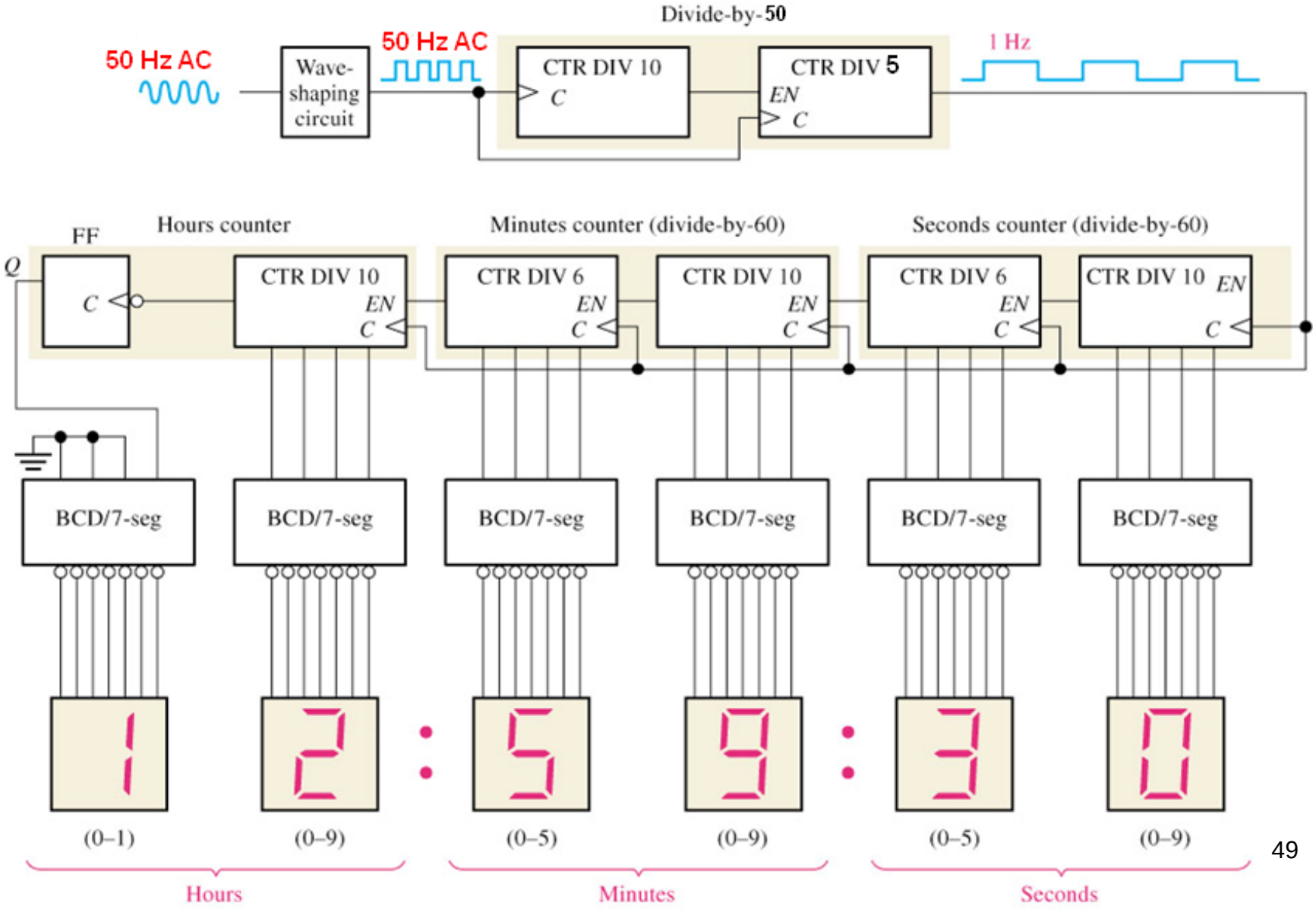
Strobing to eliminate glitches



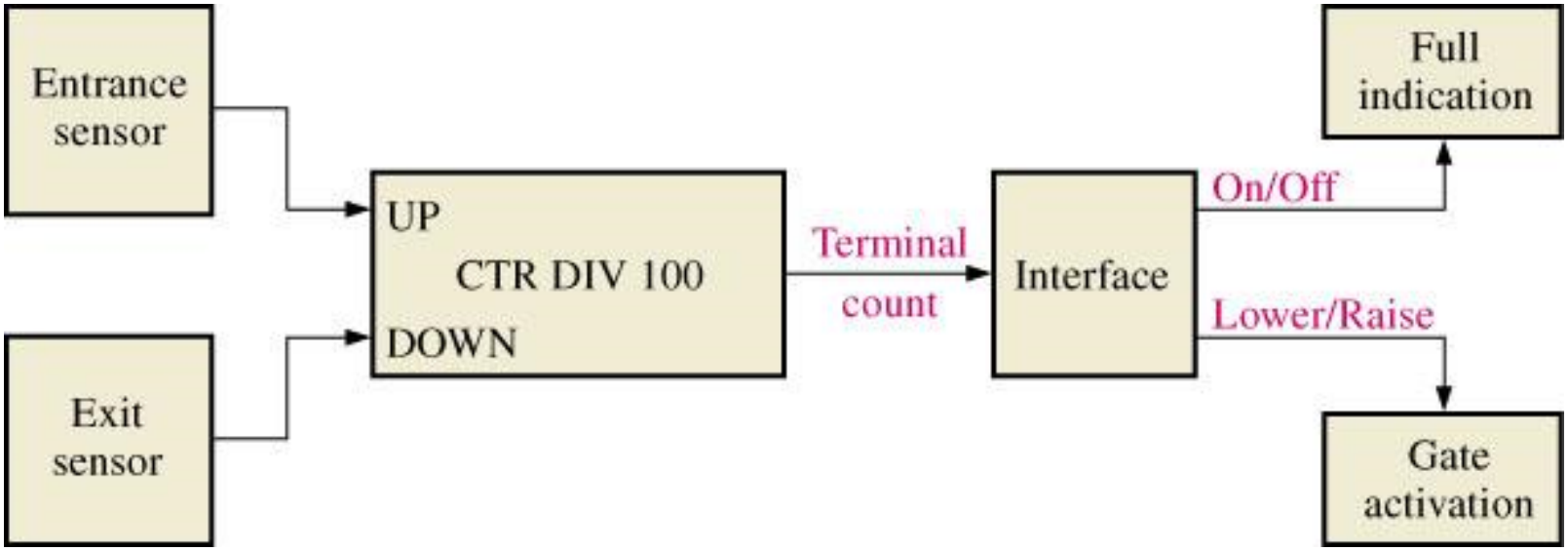


ใช้สัญญาณ Strobe เพื่อกำจัดปัญหา Glitch อันเนื่องมาจาก delay ภายในวงจรนับ

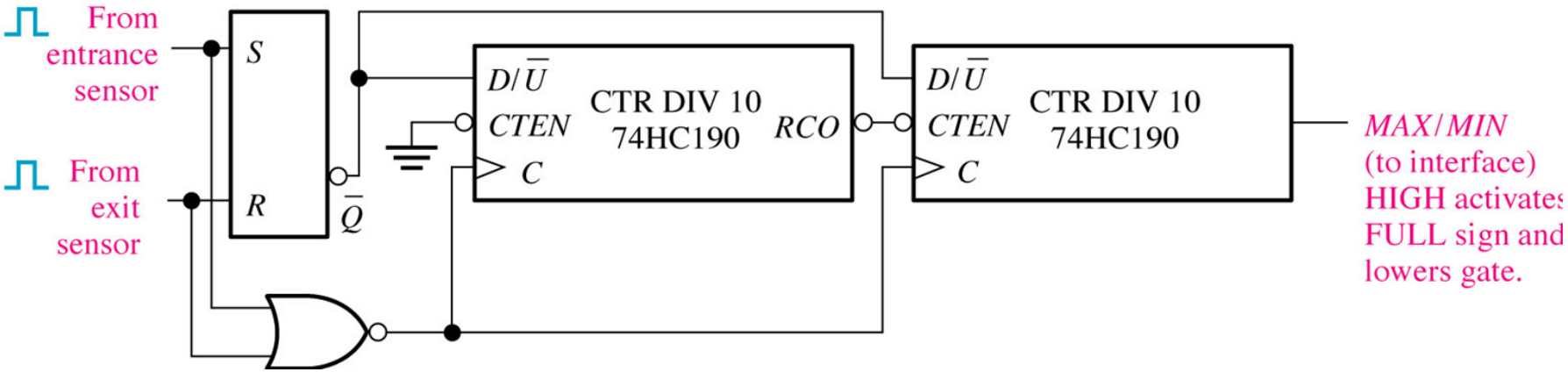
Applications : 12-hour digital clock



Applications : Automobile Parking Control

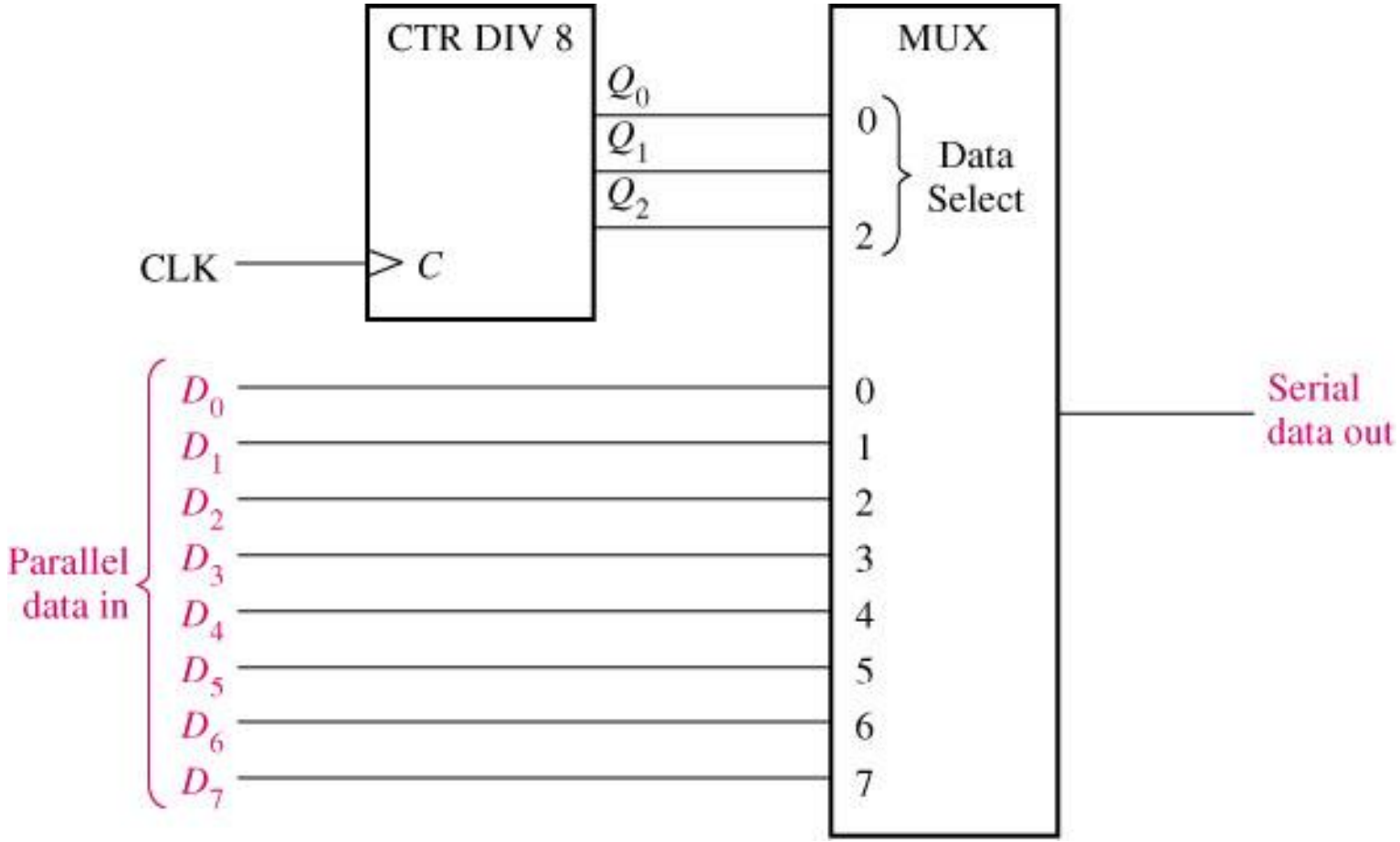


Mod-100 up/down counter for automobile parking control

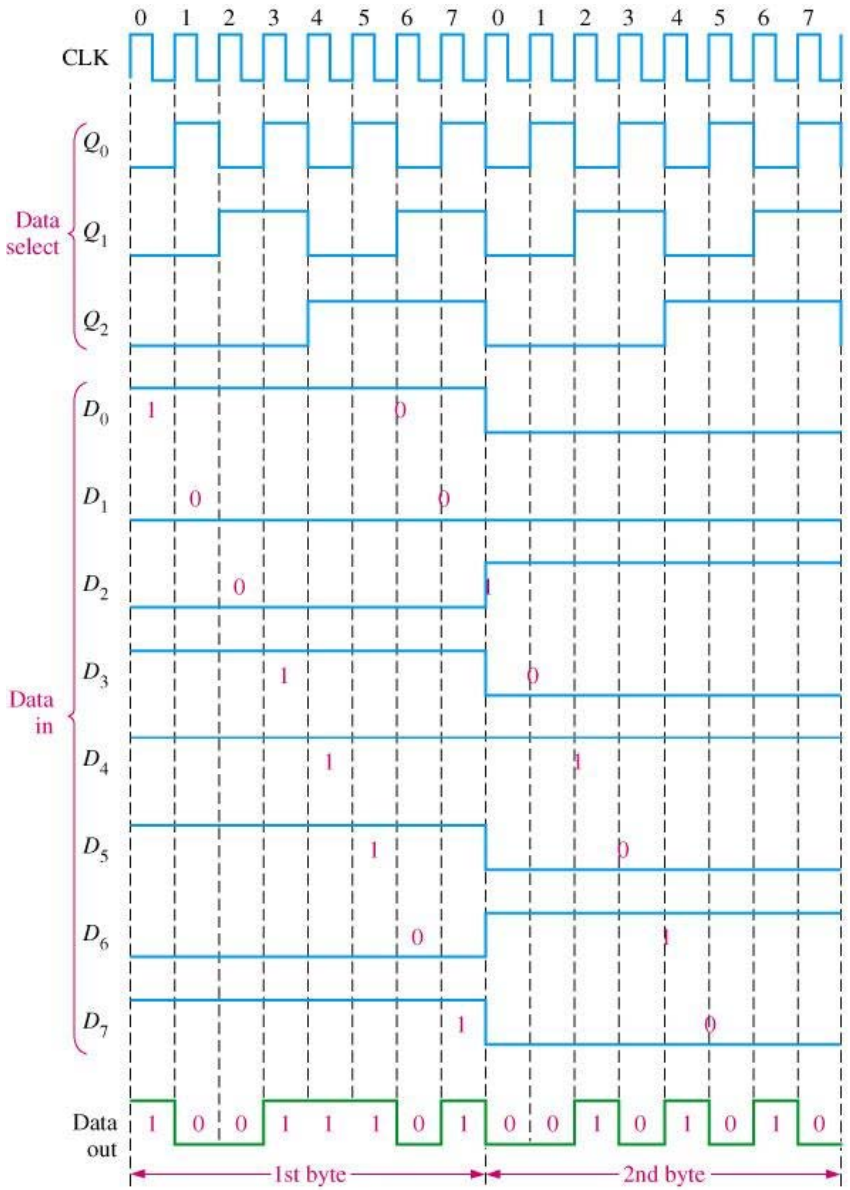


Parallel-to-serial data conversion logic

Counter Applications : Parallel-to-Serial Data Conversion (Multiplexing)

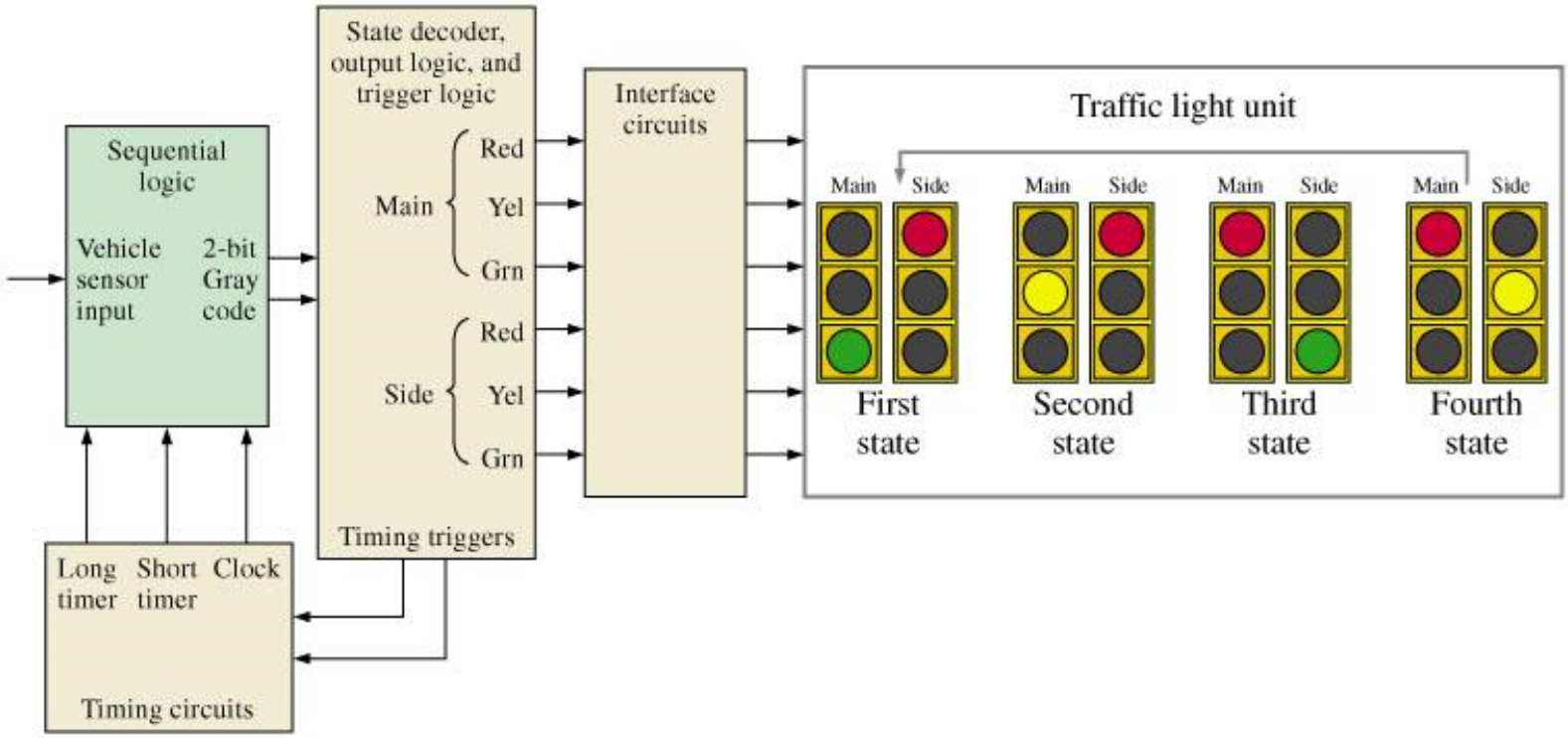


Example of parallel-to-serial conversion timing

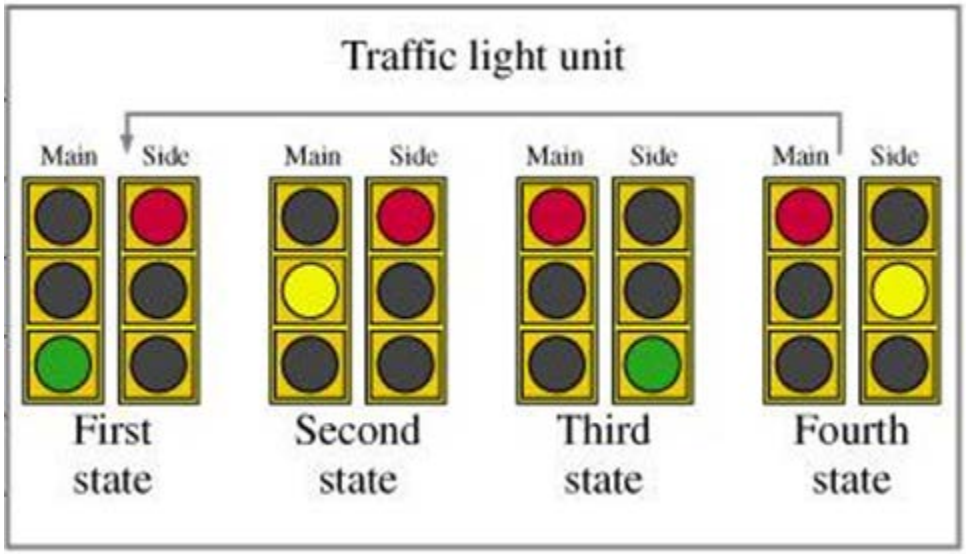


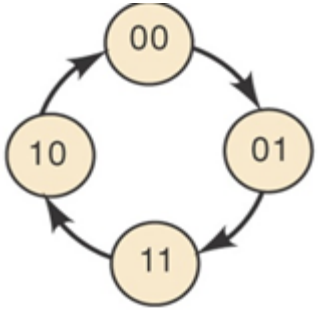
Traffic light control system block diagram

<u>การบ้าน 3</u> จงออกแบบวงจร synchronous ควบคุมสัญญาณไฟจราจรสำหรับเส้นทางตรง (main road) และเส้นทางตัด (side road) ตาม state diagram ด้านล่าง **<u>โดยใช้ JK-FF</u>**



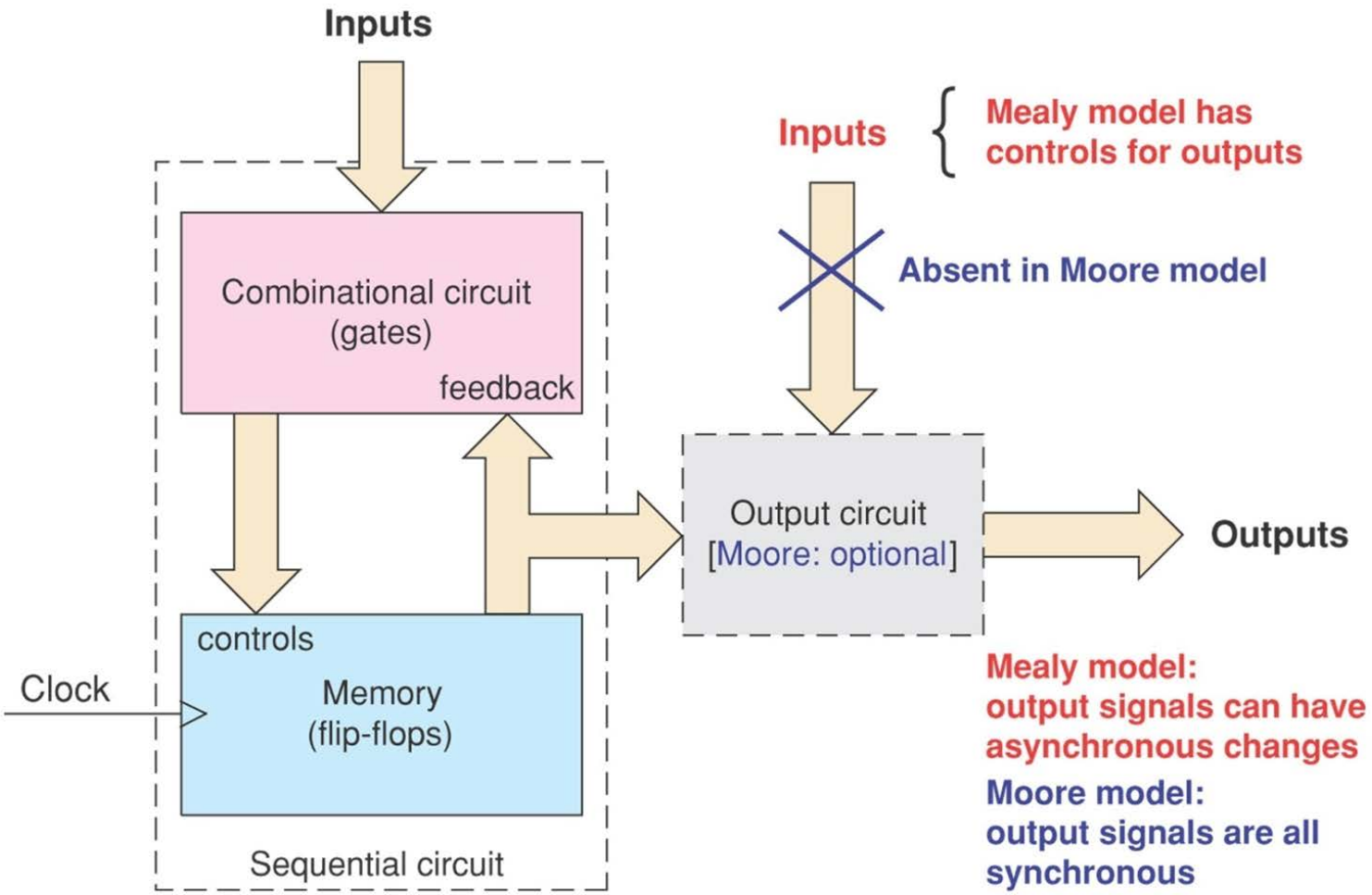
Hint: การบ้าน 3



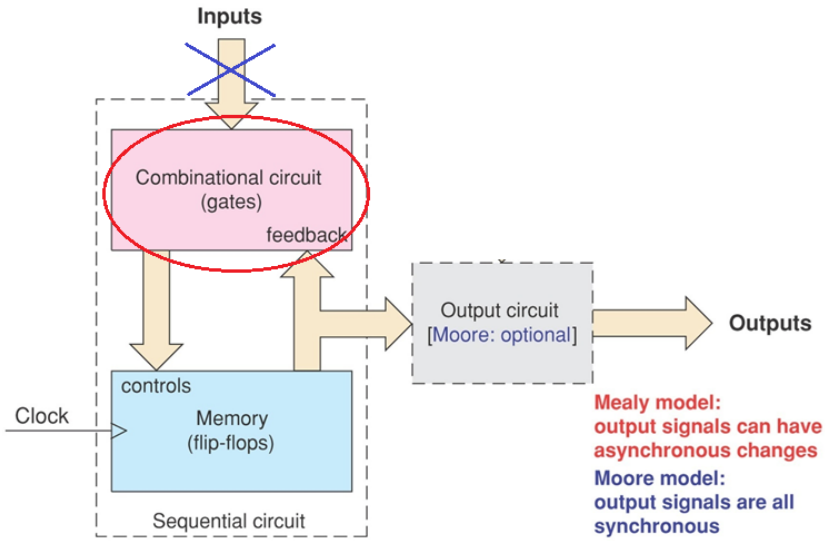


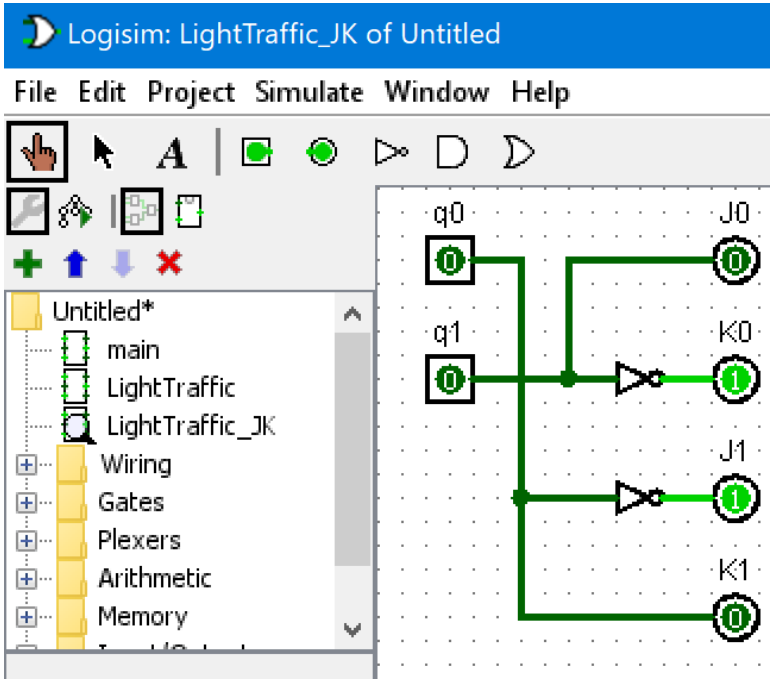
Present State		Next State	Main			Side		
			R	Y	G	R	Y	G
1st	00	01	0	0	1	1	0	0
2nd	01	11	0	1	0	1	0	0
3rd	11	10	1	0	0	0	0	1
4th	10	00	1	0	0	0	1	0

Block diagram for counters and state machines



Hint: การบ้าน 3 (ส่วน State transition)

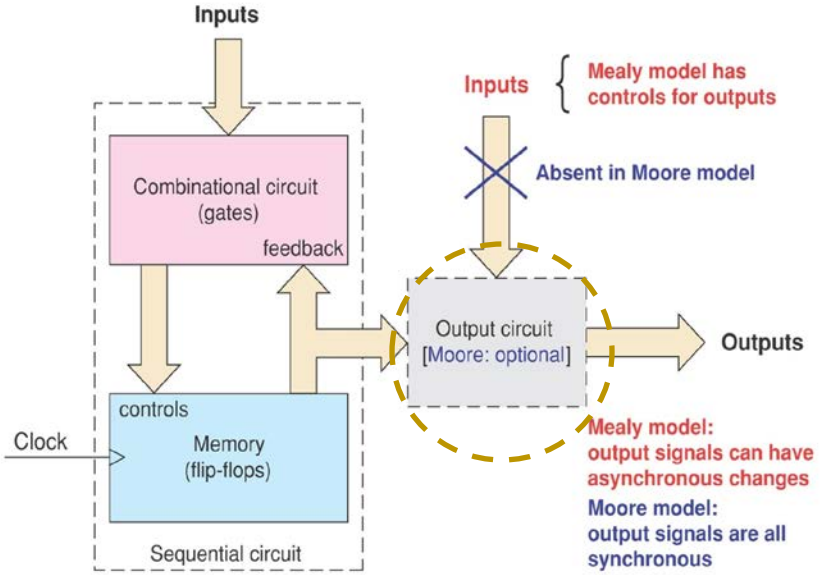




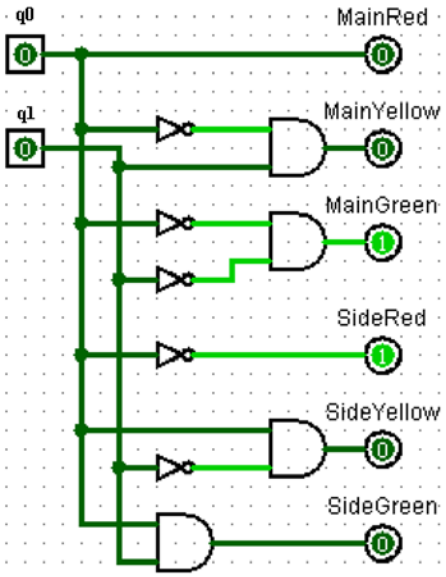
Present State		Next State	Main			Side		
	q0,q1	Q0,Q1	R	Y	G	R	Y	G
1st	00	01	0	0	1	1	0	0
2nd	01	11	0	1	0	1	0	0
4th	10	00	1	0	0	0	1	0
3rd	11	10	1	0	0	0	0	1

q0	q1	J0	K0	J1	K1
0	0	0	x	1	x
0	1	1	x	x	0
1	0	x	1	0	x
1	1	x	0	x	1

Hint: การบ้าน 3 (ส่วน Moore's Output Circuit)



Present State		Next State	Main			Side		
	q0,q1	Q0,Q1	R	Y	G	R	Y	G
1st	00	01	0	0	1	1	0	0
2nd	01	11	0	1	0	1	0	0
4th	10	00	1	0	0	0	1	0
3rd	11	10	1	0	0	0	0	1



q0	q1	MainRed	MainYellow	MainGreen	SideRed	SideYellow	SideGreen
0	0	0	0	1	1	0	0
0	1	0	1	0	1	0	0
1	0	1	0	0	0	1	0
1	1	1	0	0	0	0	1

Hint: การบ้าน 3

